

DATE : 07.10.2026
DT/NT : DT
LESSON : NETWORK
SUBJECT: OSI AND TCP/IP MODELS

BATCH : B 424-020DE

AWS-DEVOPS



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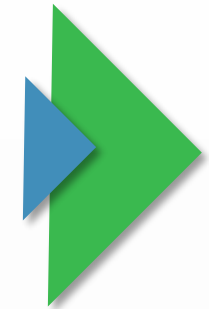


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Table of Contents

- ▶ Transmission Media
- ▶ Networking Models
 - ▶ OSI Model
 - ▶ TCP/IP Model
 - ▶ Comparison of OSI Model and TCP/IP Model



Transmission Media

Types of Network Cabling



- Coaxial
- Twisted Pair
- Fiber Optic

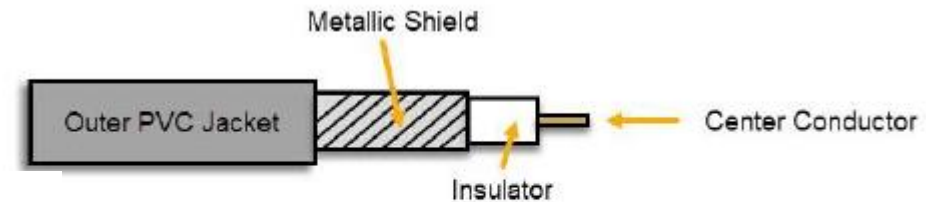
Ethernet

- Ethernet is a network protocol that controls how data is transmitted over a LAN.
- It's referred to as the Institute of Electrical and Electronics Engineers (IEEE) 802.3 Standard.
- It supports networks built with coaxial, twisted-pair, and fiber-optic cabling.
- The original Ethernet standard supported 10Mbps speeds, but the latest supports much faster gigabit speeds.
- Ethernet uses CSMA/CD & CSMA/CA access methodology.



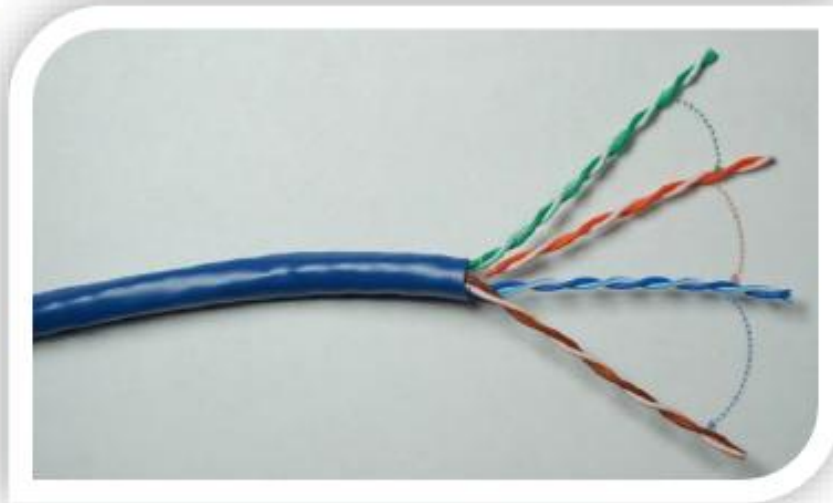
Coaxial Cable

- Antiquated technology used in the 1980s. Coaxial cables are rarely used today, except for cable modem connections.
- Categorized as Radio Grade (RG)
 - **RG-6:** Used for modern cable TV and broadband cable modems.
 - **RG-8:** Used in early 10Base5 “Thick-net” Ethernet networks.
 - **RG-58:** Used in early 10Base2 “Thin-net” Ethernet networks.
 - **RG-59:** Used for closed-circuit TV (CCTV) networks
- Metallic shield helps protect against electromagnetic interference (EMI)



Twisted Pair Copper Cabling

- 4 Twisted Pairs of Wires with RJ-45 Connector
- Balanced pair operation
 - + & - Signals
 - Equal & Opposite Signal
- Why are they twisted?
 - To Help Reduce Interference
 - Crosstalk
 - Noise (Electromagnetic Interference)
- Security concerns
 - Signal Emanations
- 100 Meters Maximum Distance
 - Signal Attenuation



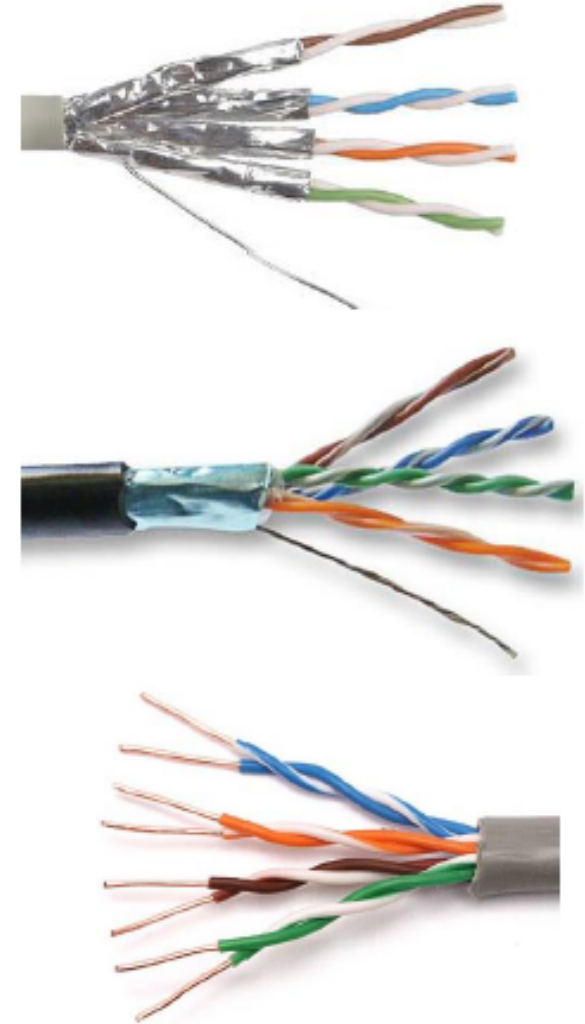
Twisted Pair Standards

Cat	Network Type	Ethernet Standard	Speed	Max. Distance
Cat 3	Ethernet	10Base-T	10Mbps	100 meters
Cat 5	Fast Ethernet	100Base-TX	100Mbps	100 meters
Cat 5e	Gigabit Ethernet	1000Base-T	1Gbps	100 meters
Cat 6	Gigabit Ethernet	1000Base-T	1Gbps	100 meters
	10 Gigabit Ethernet	10GBase-T	10Gbps	55 meters
Cat 6a	10 Gigabit Ethernet	10GBase-T	10Gbps	100 meters
Cat 7	10 Gigabit Ethernet	10GBase-T	10Gbps	100 meters

Cat: Copper Cabling Standard.

Shielded vs. Unshielded & EMI

- **Unshielded Twisted Pair (UTP)**
 - More susceptible to electromagnetic interference (EMI).
- **Shielded Twisted Pair (STP)**
 - Less susceptible to EMI & Crosstalk (if each pair shielded).
- **Electromagnetic Interference**
 - The disruption of an electronic device's operation when it's in the vicinity of an electromagnetic field caused by another electronic device (manufacturing equipment, microwave ovens, etc.).



Straight-Through & Crossover Cables

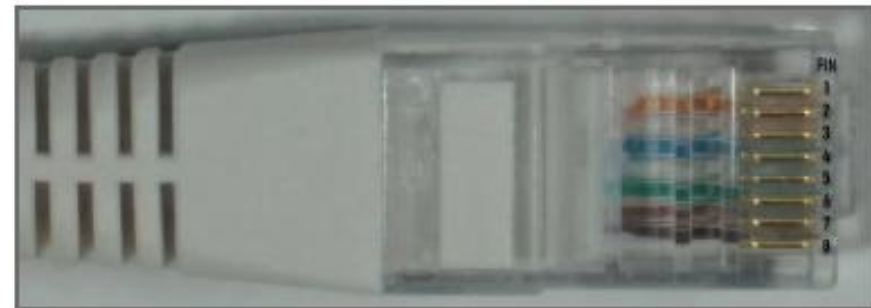
Straight-Through Cable

- Connecting “Unlike” Devices
 - Computer to Switch
 - Switch to Router



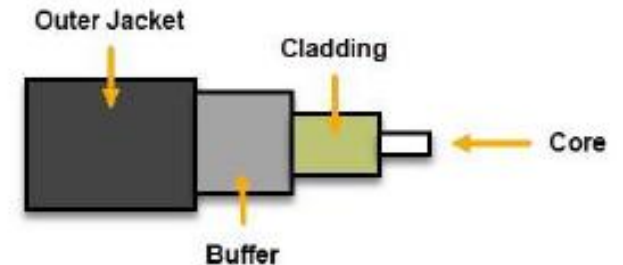
Crossover Cable

- Connecting “Like” Devices
 - Router to Router
 - Computer to Computer



Fiber Optic Cabling

- Glass or plastic fiber that carries light (photons)
 - **High Bandwidth:** Photons travel faster than electrons.
 - **Long Distances:** Less attenuation.
 - Immune to Electromagnetic Interference (EMI)
 - Doesn't Emanate Signals
- Two Types
 - **Multi-mode Fiber (MMF)**
 - Shorter Distances (LAN / Building-to-Building)
 - Up to 2 Kilometers
 - **Single-mode Fiber (SMF)**
 - More expensive than multi-mode
 - Longer Distances (WAN / Across Town)
 - Up to 200 Kilometers

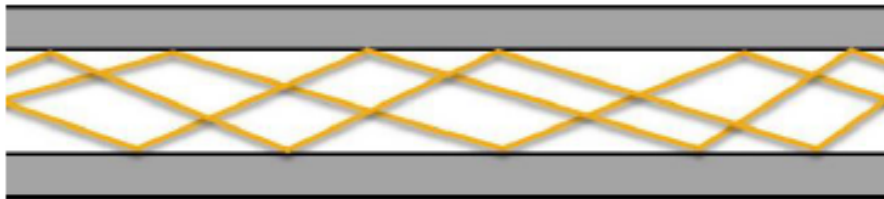


Informational Note: 9-micron Single-Mode Fiber can travel 75 miles at 400 Gbps

MMF versus SMF

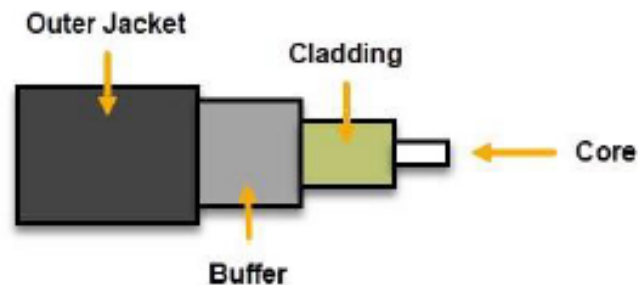
Multi-Mode Fiber (MMF)

- Many photons of light travel through the cable at once, and bounce off the walls, which reduces the distance and speed.
- **Larger Core:** 50 to 62.5 microns



Single-Mode Fiber (SMF)

- A single direct photon of light travels through the cable, which allows greater distances and speed.
- **Smaller Core:** 8 to 10 microns



Why use Fiber?

- Fiber cable is more expensive than twisted pair, as is the equipment
- But you can perform much longer network cable runs with fiber.
 - 100m versus up to 200 Kilometers
- So you have decreased network equipment costs
 - Switches, routers, etc.
- Plus fiber is:
 - Immune to EMI and signal emanations
 - Has lower signal attenuation
 - Making it more reliable and secure
- Costs are steadily decreasing as more people adopt fiber

Cable Selection Criteria

Cost Constraints

- What is your budget?

Transmission Speed Requirements

- How fast does your network need to be?
- 10Mbps, 100Mbps, 1Gbps, 10Gbps?

Distance Requirements

- Electrical signals degrade relatively quickly (100 meters)
- Fiber can transmit over long distances

Noise & Interference Immunity (Crosstalk, EMI, Security)

- Interference is all around us: power cables, microwaves, mobile phones, motors, etc.



Networking Models

- ▶ **OSI Model**
- ▶ **TCP/IP Model**
- ▶ **Comparison of OSI Model and TCP/IP Model**

NETWORKING MODELS

OSI Model is a generic network model that may describe how an ideal network would behave & function

On the other hand TCP/IP is implementation of OSI in a specific case of internet. TCP/IP model outlines interconnection between the network devices on 

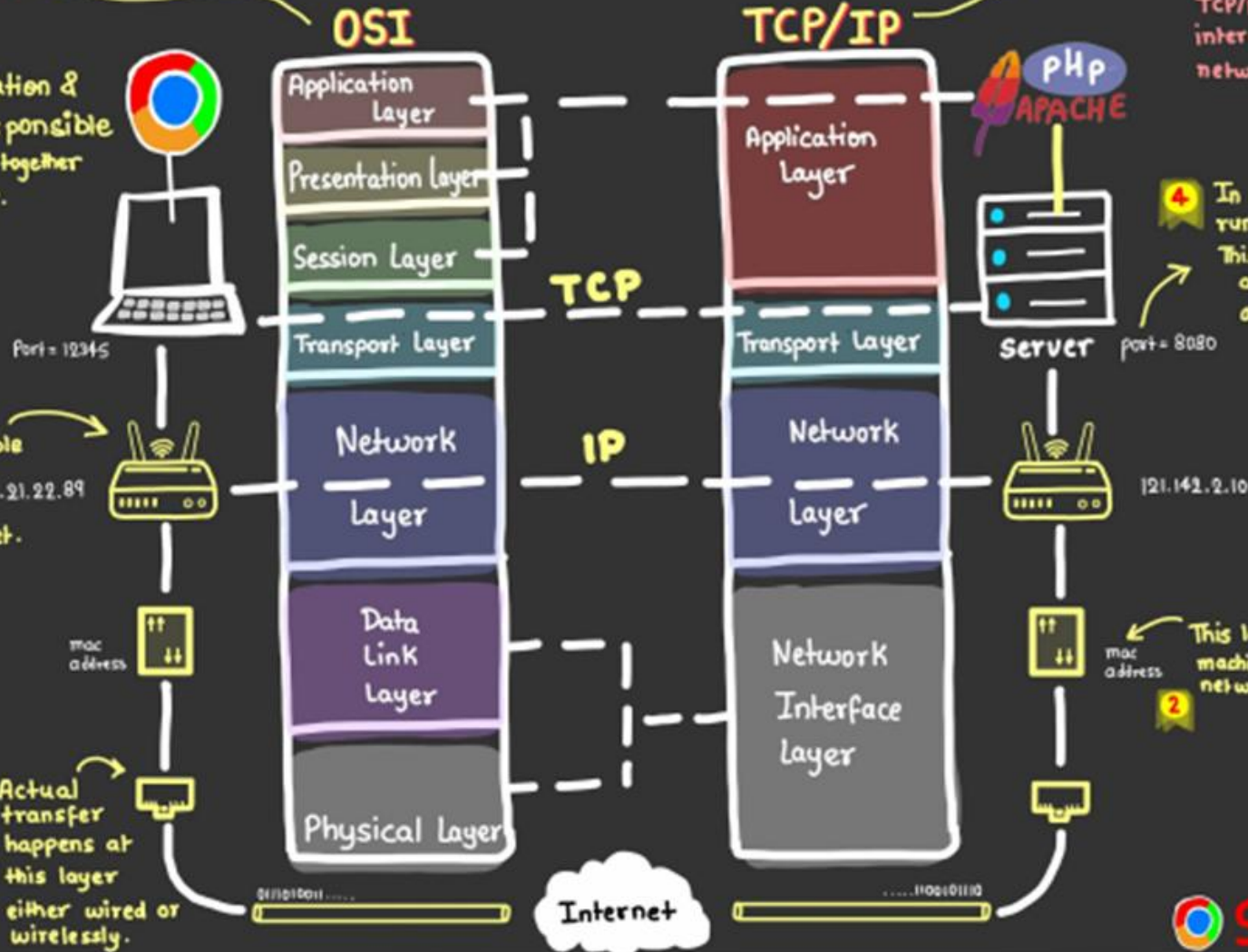
5 Application, presentation & session layer are responsible for binding everything together & showing that to user.

3 This layer is responsible for packet delivery across networks e.g delivery over internet.

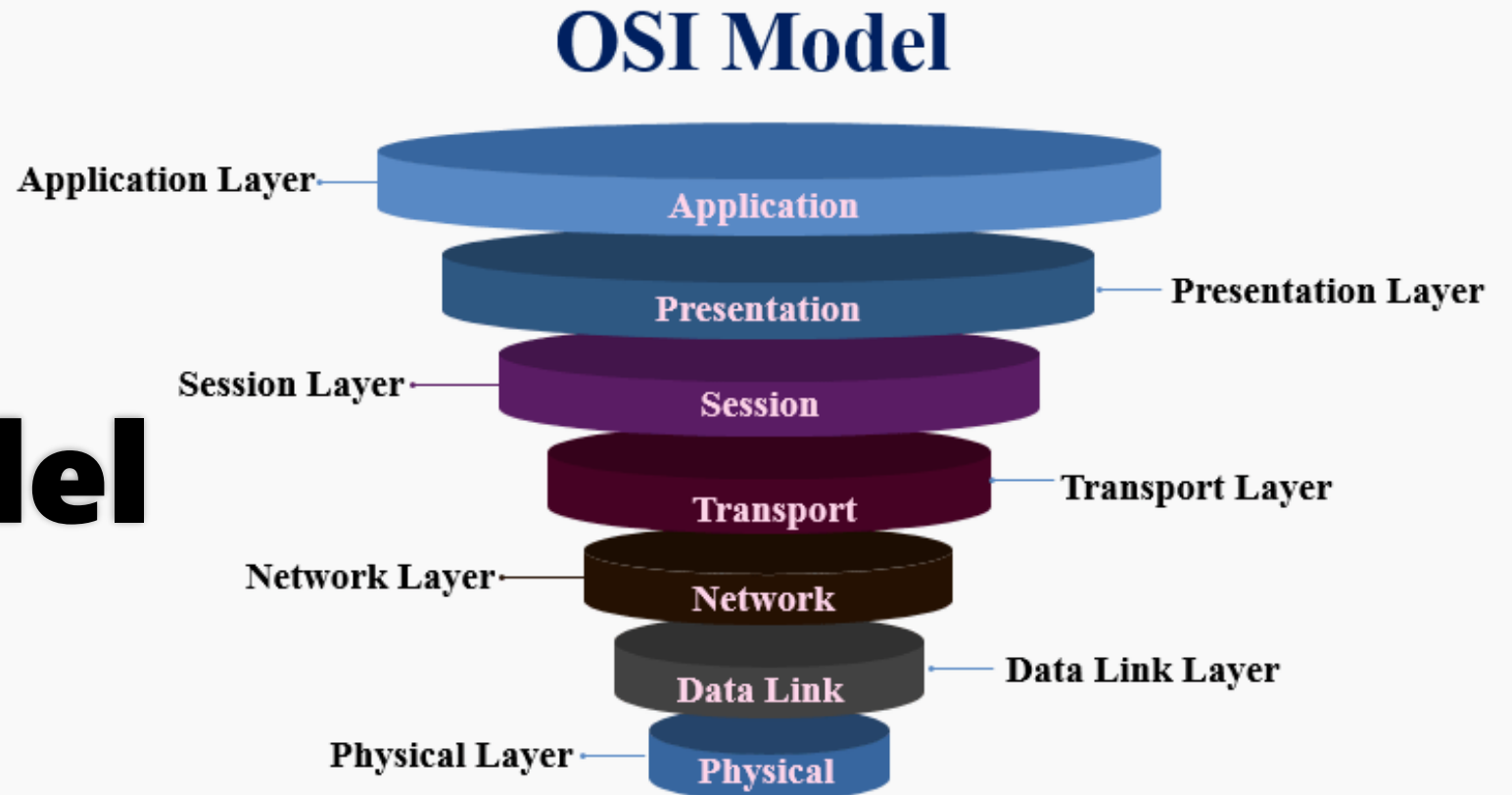
1 Actual transfer happens at this layer either wired or wirelessly.

4 In a single machine there runs multiple application. This layer identifies that application via port number and handles packet to that.

2 This layer is responsible for machine identification in local network.

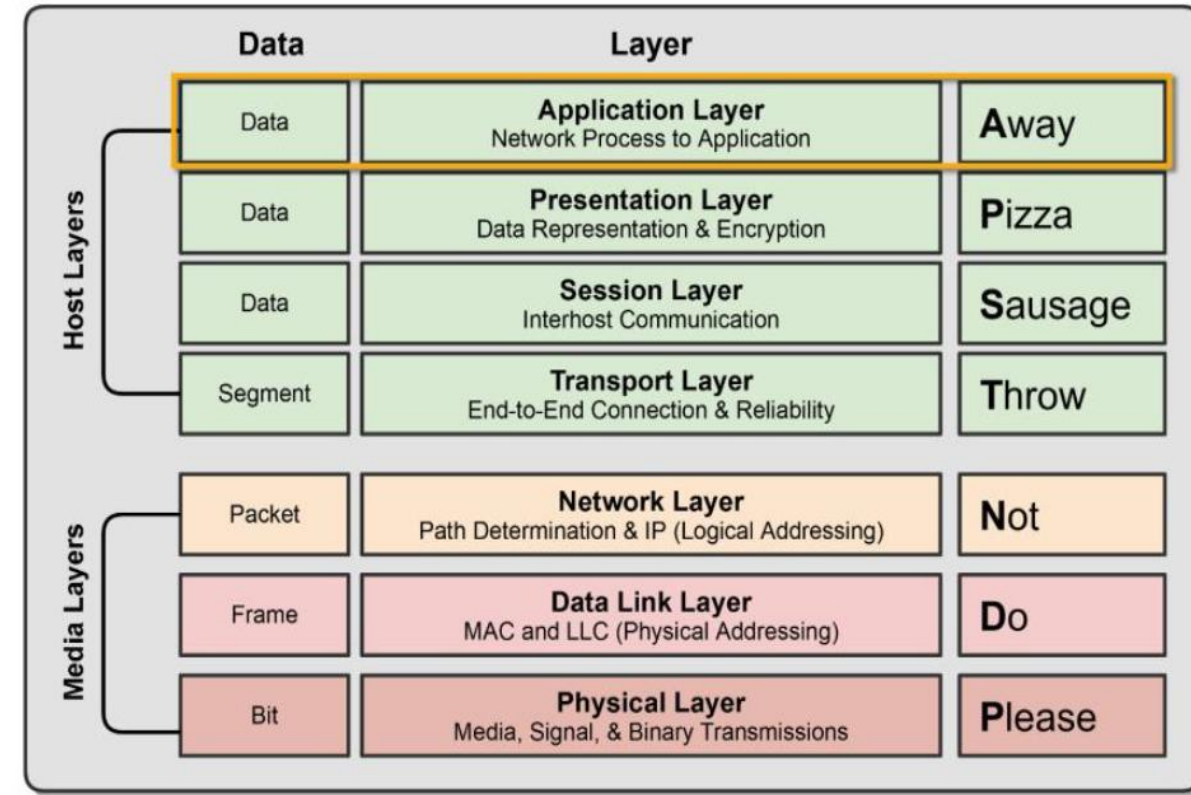


OSI Model



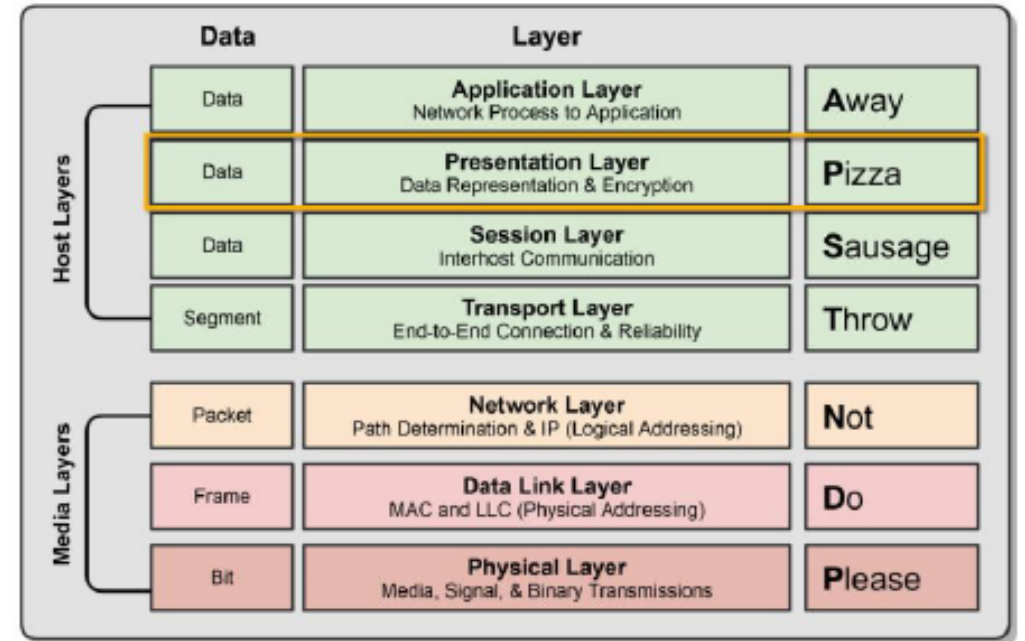
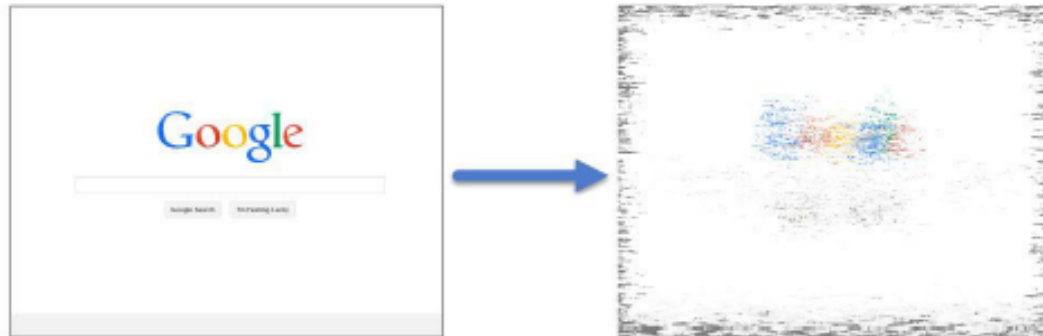
Layer 7 – Application Layer

- Where users interact with the computer.
- Acts as an interface between an application and end-user protocols.
- Provides an interface to communicate with the network (Outlook, Chrome, etc.).
- Applications don't reside in the application layer but instead interfaces with application-layer protocols.
- Example Application Layer Protocols:
 - **E-Mail:** IMAP4, POP3, SMTP
 - **Web Browsers:** HTTP, HTTPS
 - **Remote Access:** SSH, Telnet



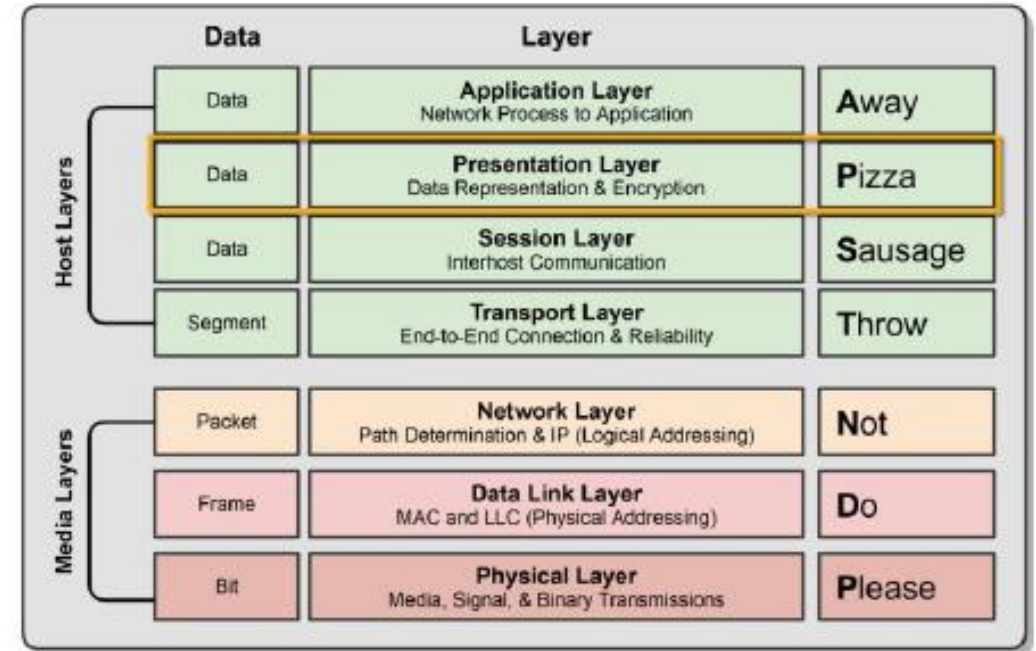
Layer 6 – Presentation Layer

- Ensures that data transferred from one system's Application Layer can be read by the Application Layer on another one.
- Provides character code conversion, data compression, and data encryption/decryption.
- Example:** Google Chrome HTML converted to ASCII Format.



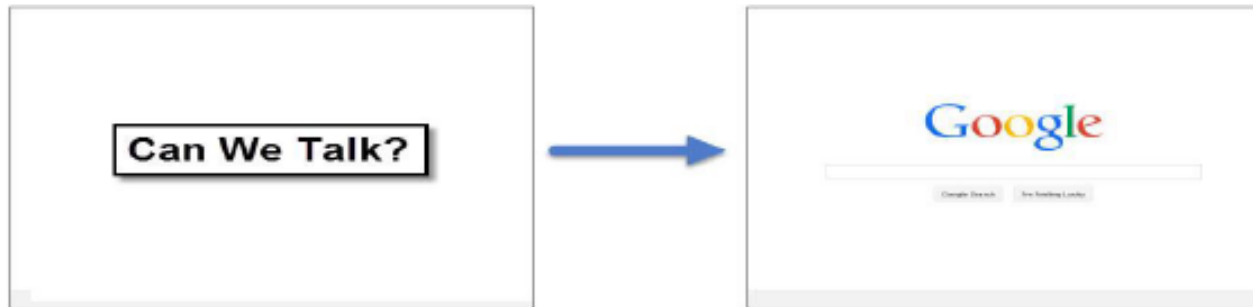
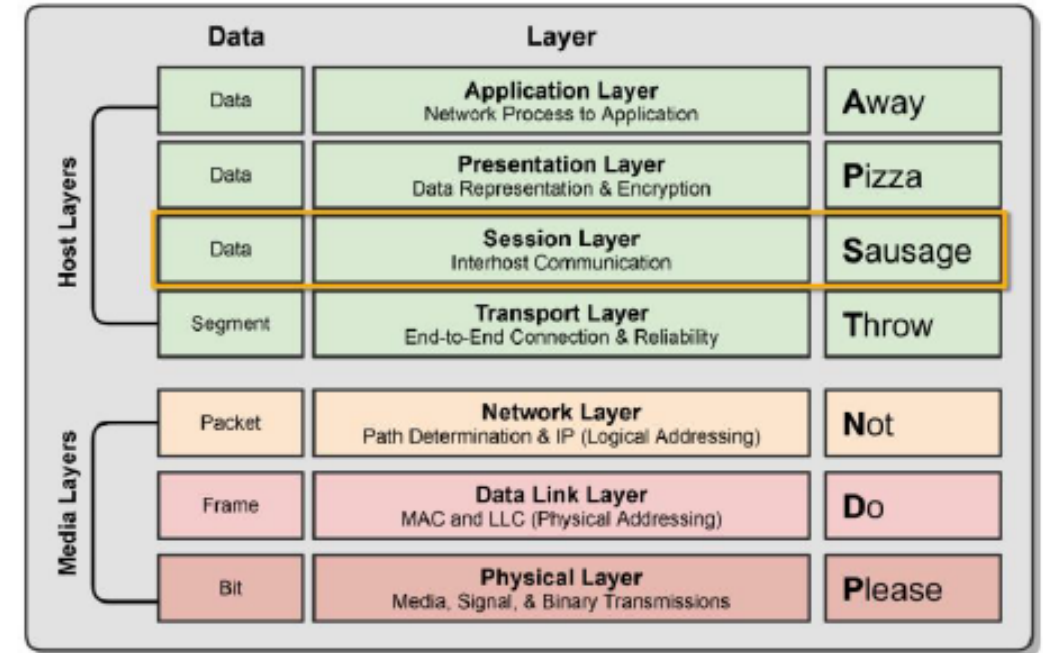
Layer 6 – Presentation Layer

- Example Layer 6 File Formats:
 - **Web Browser:** HTML, XML, JavaScript
 - **Graphics Files:** JPEG, GIF, PNG
 - **Audio/Video:** MPEG, MP3
 - **Encryption:** TLS, SSL
 - **Text/Data:** ASCII, EBCDIC



Layer 5 - Session Layer

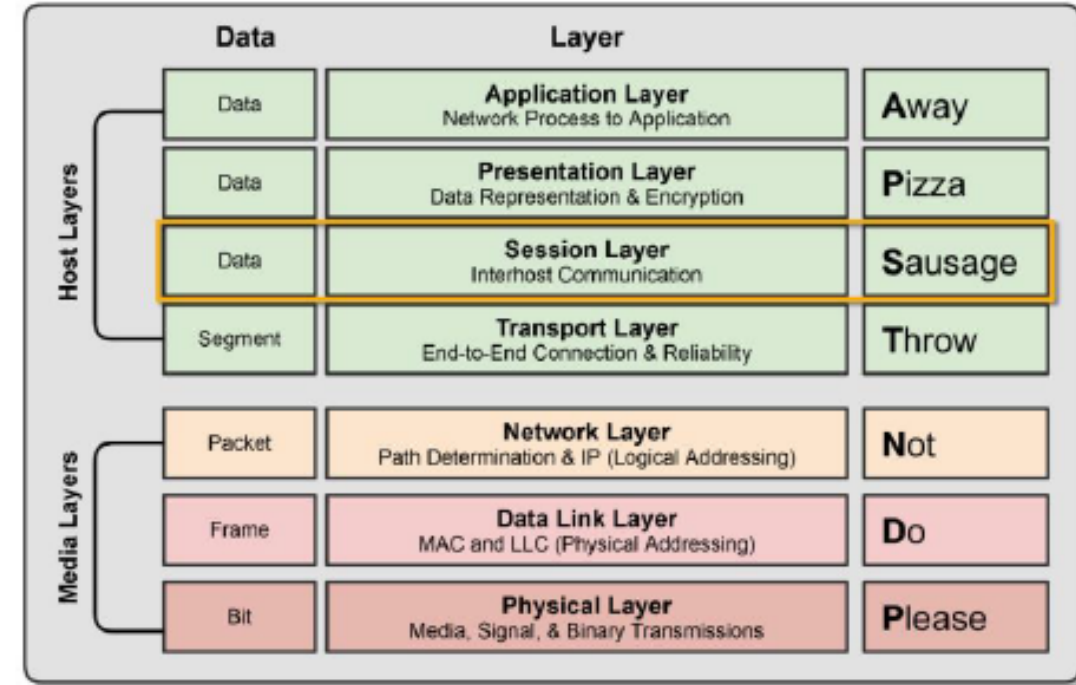
- Responsible for setting up, managing, and then tearing down sessions between network devices.
- Ensures data from different application sessions are kept separate.
- Utilizes Application Program Interfaces (APIs) to communicate with TCP/IP protocols.
- Coordinates communication between systems.
 - Start, Stop, Restart



Layer 5 - Session Layer

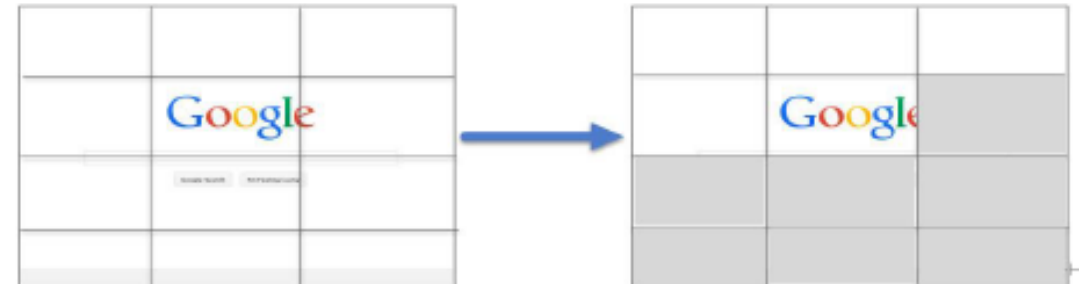
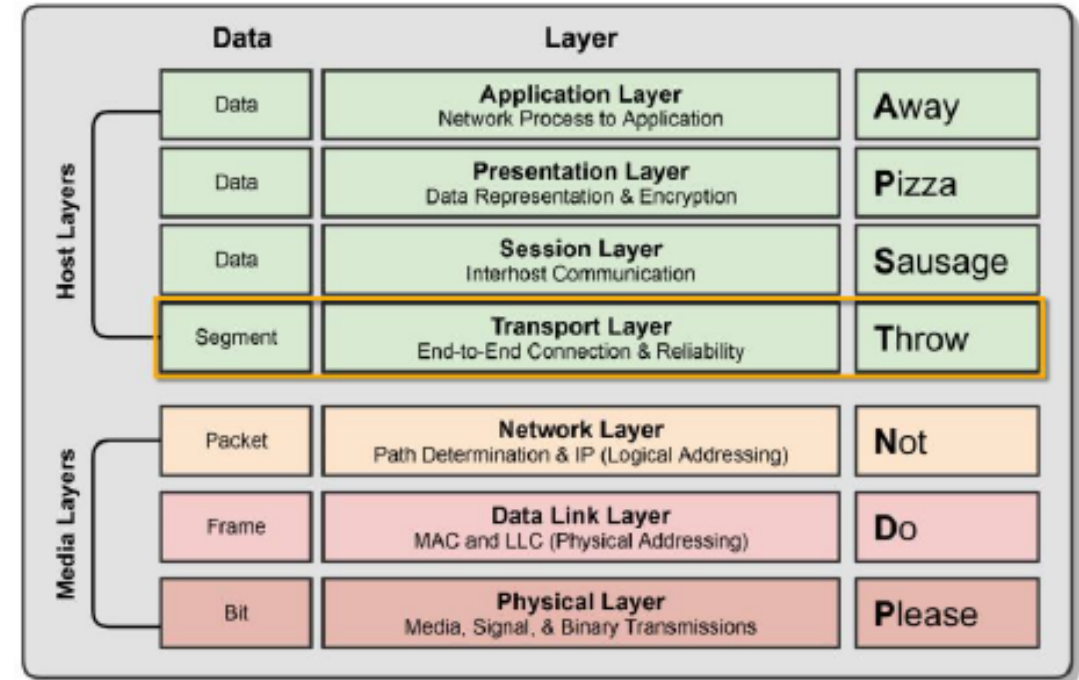
Can provide three different methods of communication between devices:

- **Simplex:** One-way communication between two devices, like listening to a radio station.
- **Half Duplex:** Two-way communication between two devices, but only one device can communicate at a time.
- **Full Duplex:** Two-way communication between two devices, where both sides can communicate at the same time.



Layer 4 - Transport Layer

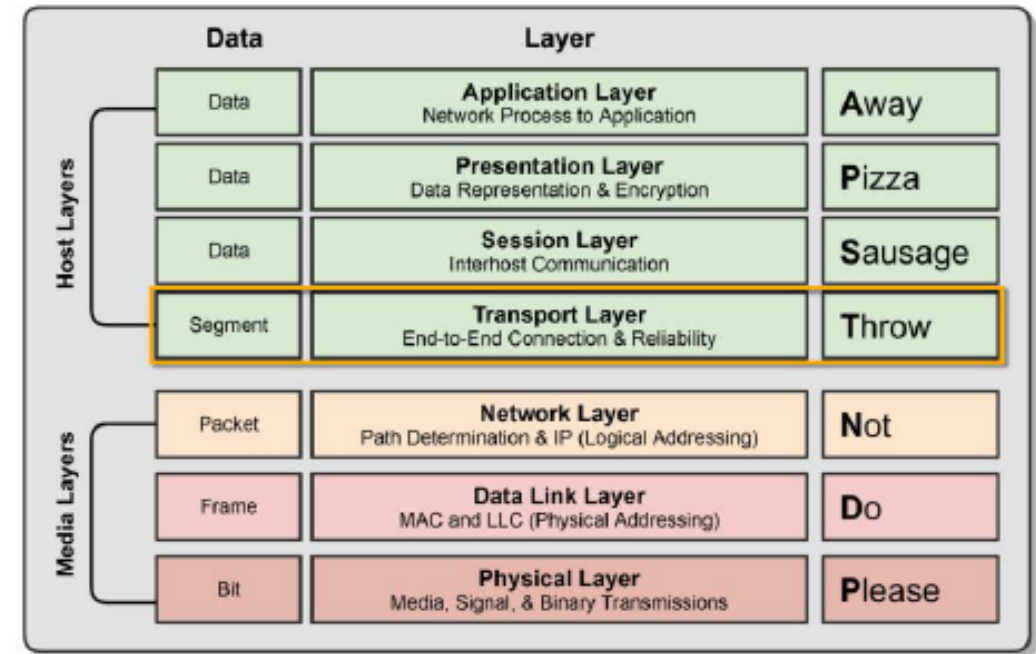
- Ensures data is delivered error-free and in sequence.
- Segments data and reassembles correctly.
- Can be connection-oriented or connectionless.
- Considered the “Post Office” Layer
 - **TCP** (Transmission Control Protocol)
 - **UDP** (User Datagram Protocol)
 - Covered in detail in the next section.



Layer 4 - Transport Layer

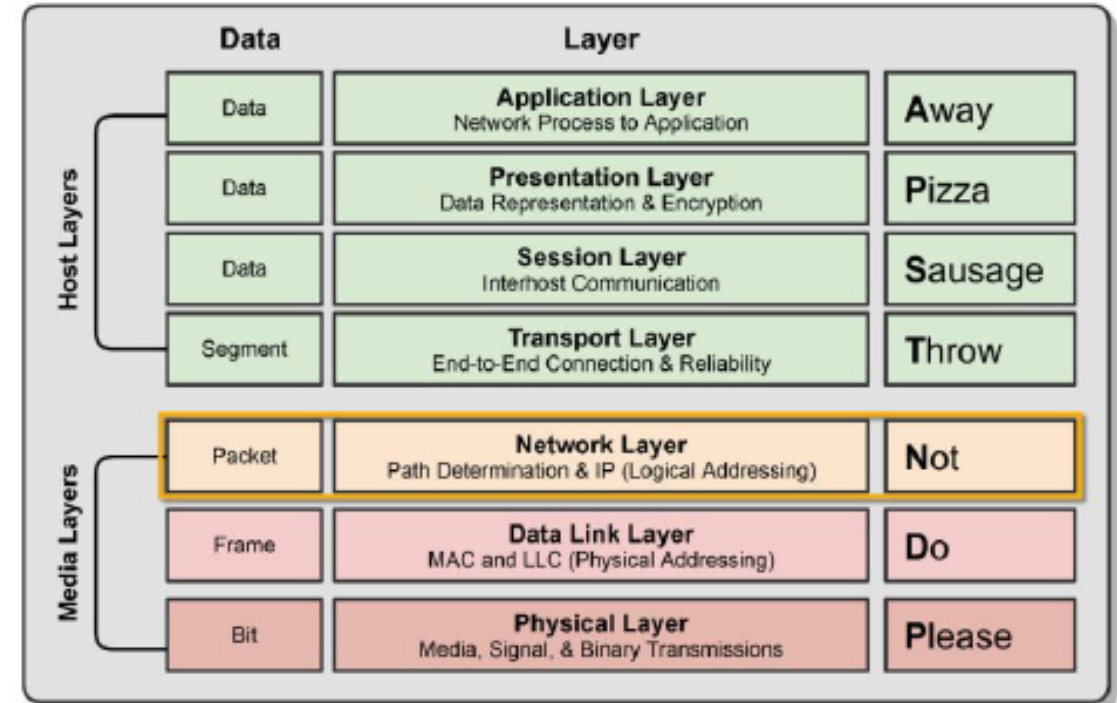
Responsible for two data flow control measures:

- **Buffering**
 - Stores data in memory buffers until destination device is available.
- **Windowing**
 - Allows devices in session to determine the “window” size of data segments sent.



Layer 3 - Network Layer

- The “Routing” Layer
- Provides logical addressing (IP Addressing) and routing services.
- Places two IP addresses into a packet:
 - Source Address & Destination IP Address
- Internet Protocol (IP)
 - The primary network protocol used on the Internet, IPv4, IPv6 Logical Addresses



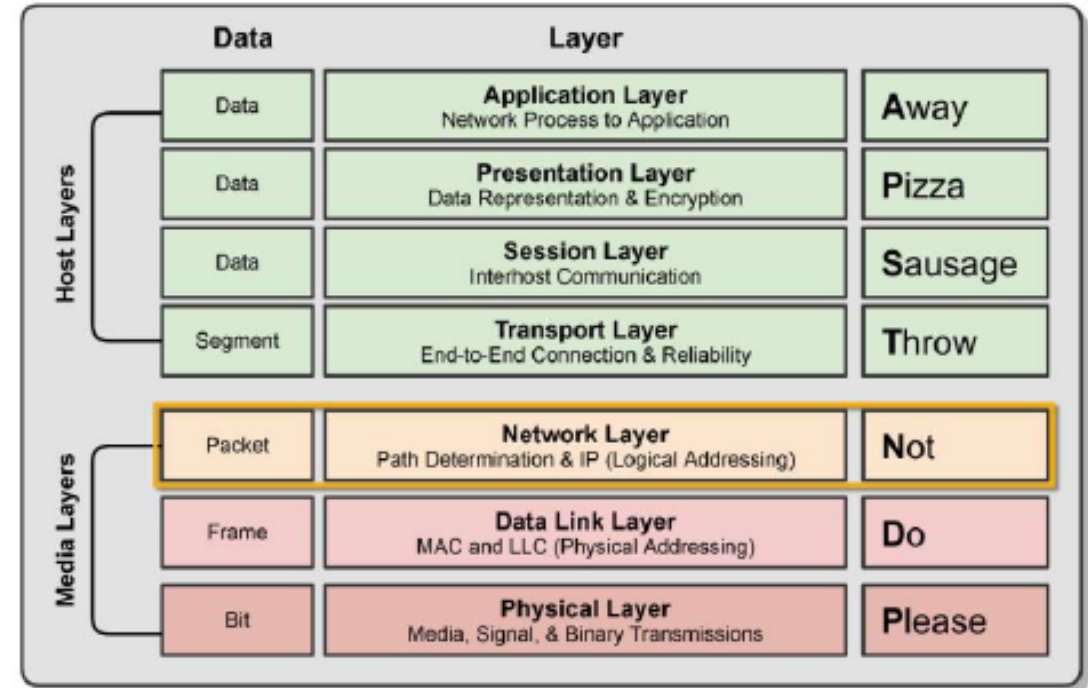
Layer 3 - Network Layer

Types of Packets at Network Layer:

- **Data Packets**
 - Routed Internet Protocol (IP) data packets.
 - IPv4 & IPv6
- **Route-Update Packets**
 - Routing protocols designed to update neighboring routers with router information for path determination.
 - RIP, OSPF, EIGRP, etc.

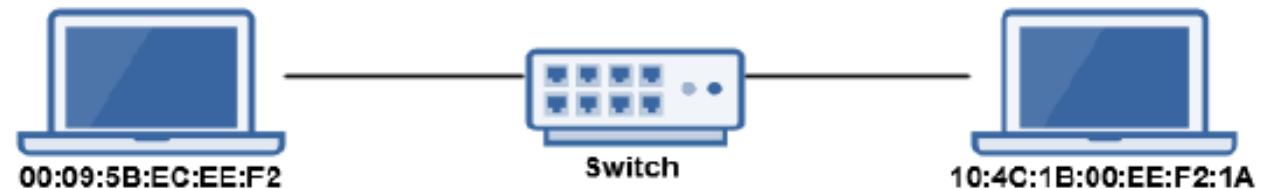
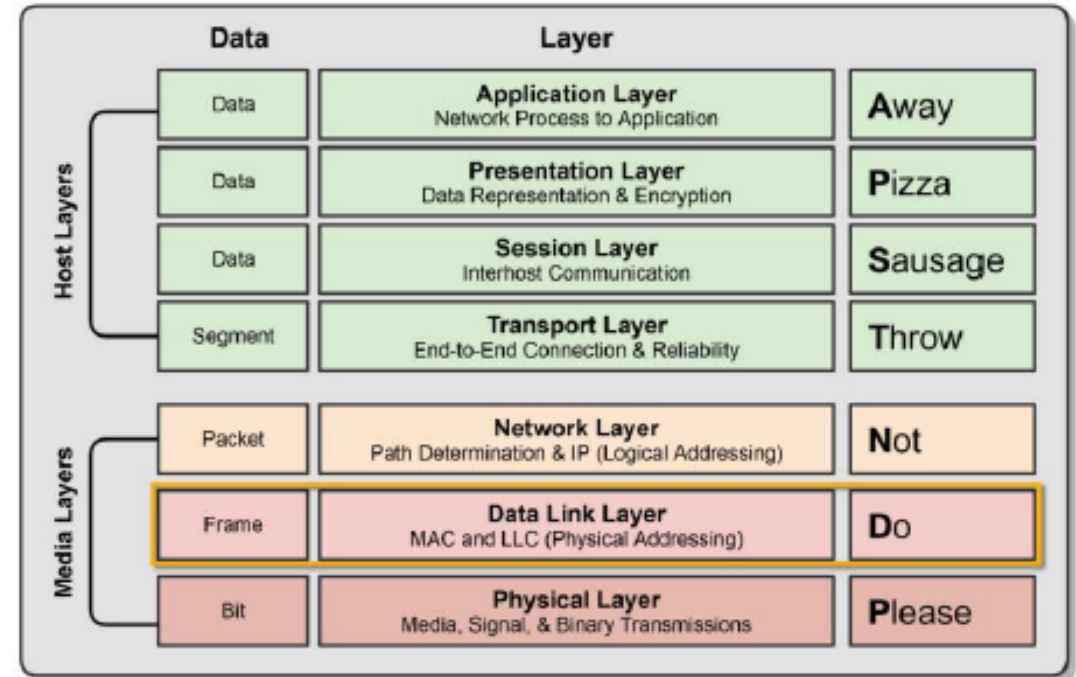
Layer 3 Devices & Protocols:

- Routers & Multi-Layer Switches
- IPv4 & IPv6
- Internet Control Message Protocol (ICMP), i.e., Ping



Layer 2 – Data Link Layer

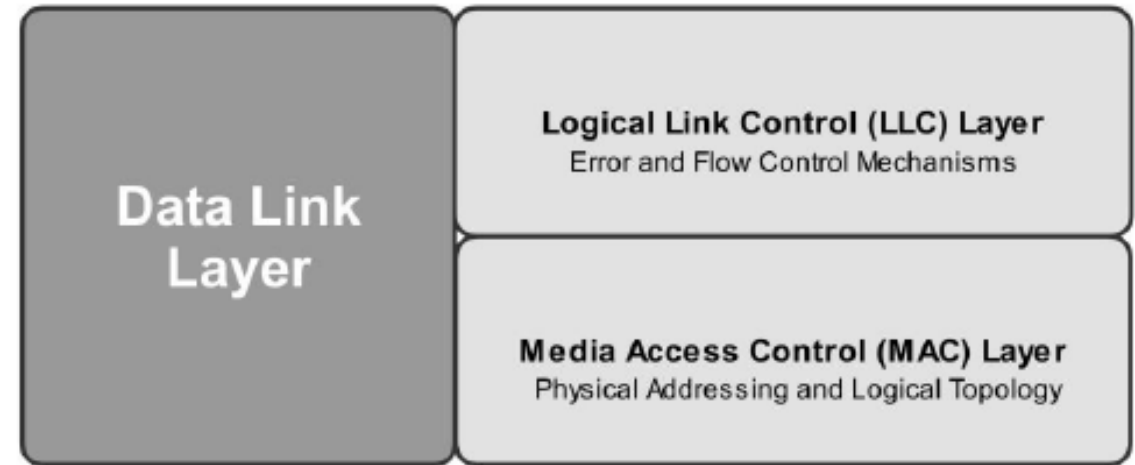
- The “Switching” Layer
- Ensures that messages are delivered to the proper device on a LAN using hardware addresses.
 - MAC (Media Access Control) Address
 - Only concerned with the local delivery of frames on the same network.
- Responsible for packaging the data into frames for the physical layer.
- Translates messages from the Network layer into bits for the Physical layer.



Layer 2 – Data Link Layer

Has two Sub-Layers

- **Logical Link Control (LLC) Layer**
 - Error Control and Flow Control
 - Detect and correct corrupted data frames.
 - Limits amount of data sent so devices aren't overwhelmed.
- **Media Access Control (MAC) Layer**
 - Physical Addressing (MAC Address)
 - 48-Bit MAC Address burned on NIC.
 - Logical Topology and Media Access
 - Ethernet, Token Ring, etc.
 - CSMA/CD & CSMA/CA

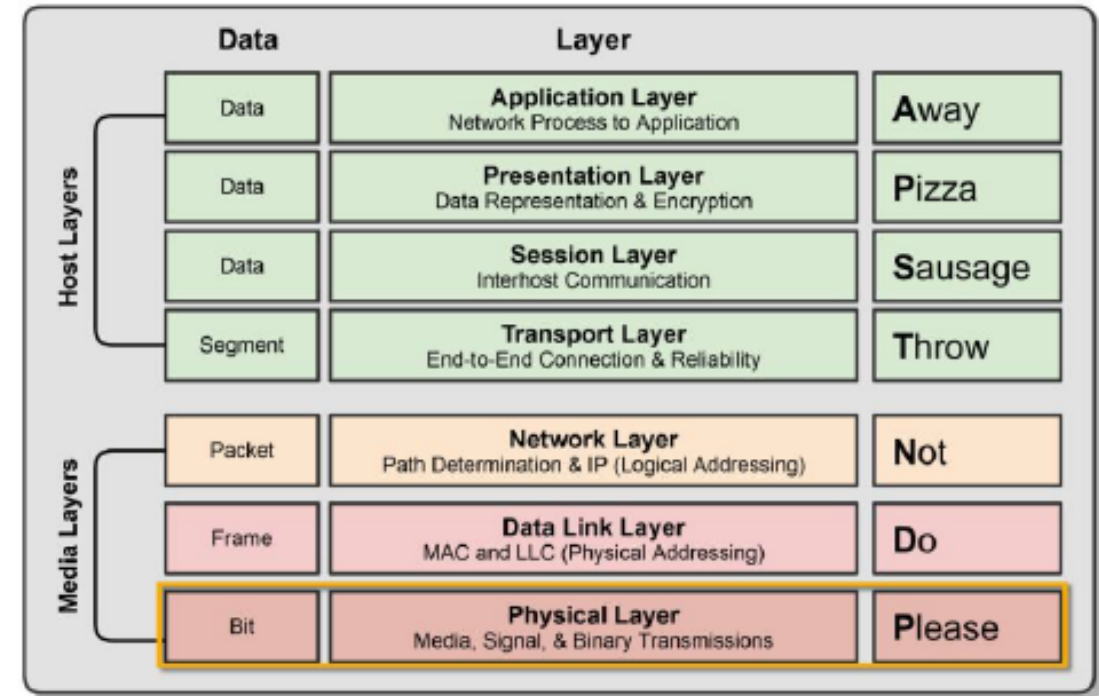


Layer 1 – Physical Layer

Defines the physical and electrical medium for network communication:

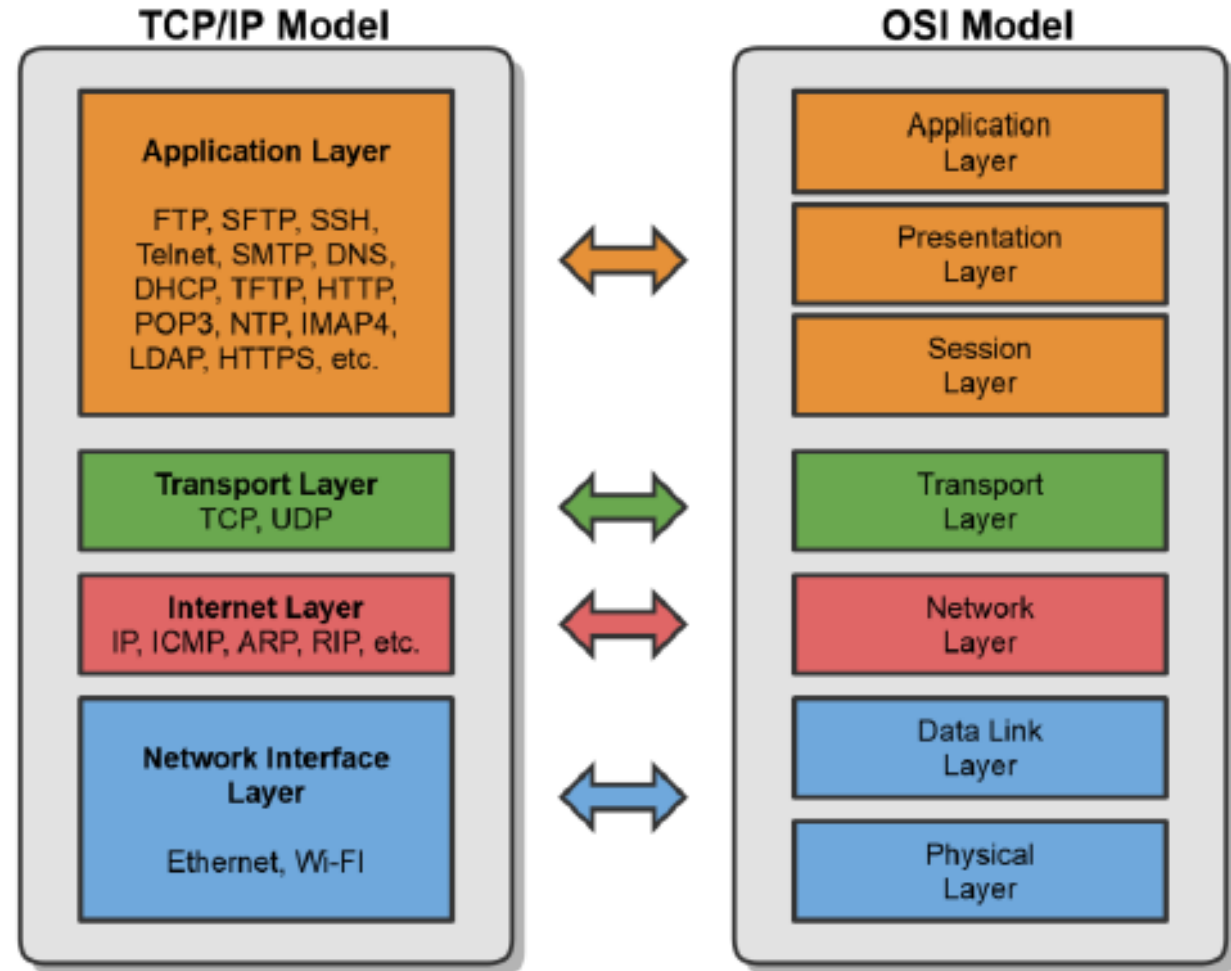
- Sending and receiving bits (1 or 0)
- Encoding Signal Types
 - Electricity, radio waves, light
- Network Cabling, Jacks, Patch Panels, etc.
 - Copper or Fiber
- Physical Network Topology
 - Star, Mesh, Ring, etc.
- Ethernet IEEE 802.3 Standard
- Layer 1 Equipment
 - Hubs, Media Converters, Modems

It's responsible for the network hardware and physical topology.



What is TCP/IP?

- Transmission Control Protocol/Internet Protocol
- Commonly called the **Internet Protocol** suite because it was designed for the Internet, but LANs use it too.
- First Two Protocols Defined in the Suite Were:
 - TCP & IP, hence TCP/IP
- Similar to the OSI Model, but Simpler:
 - OSI is Conceptual
 - TCP/IP was Implemented



TCP/IP Protocols

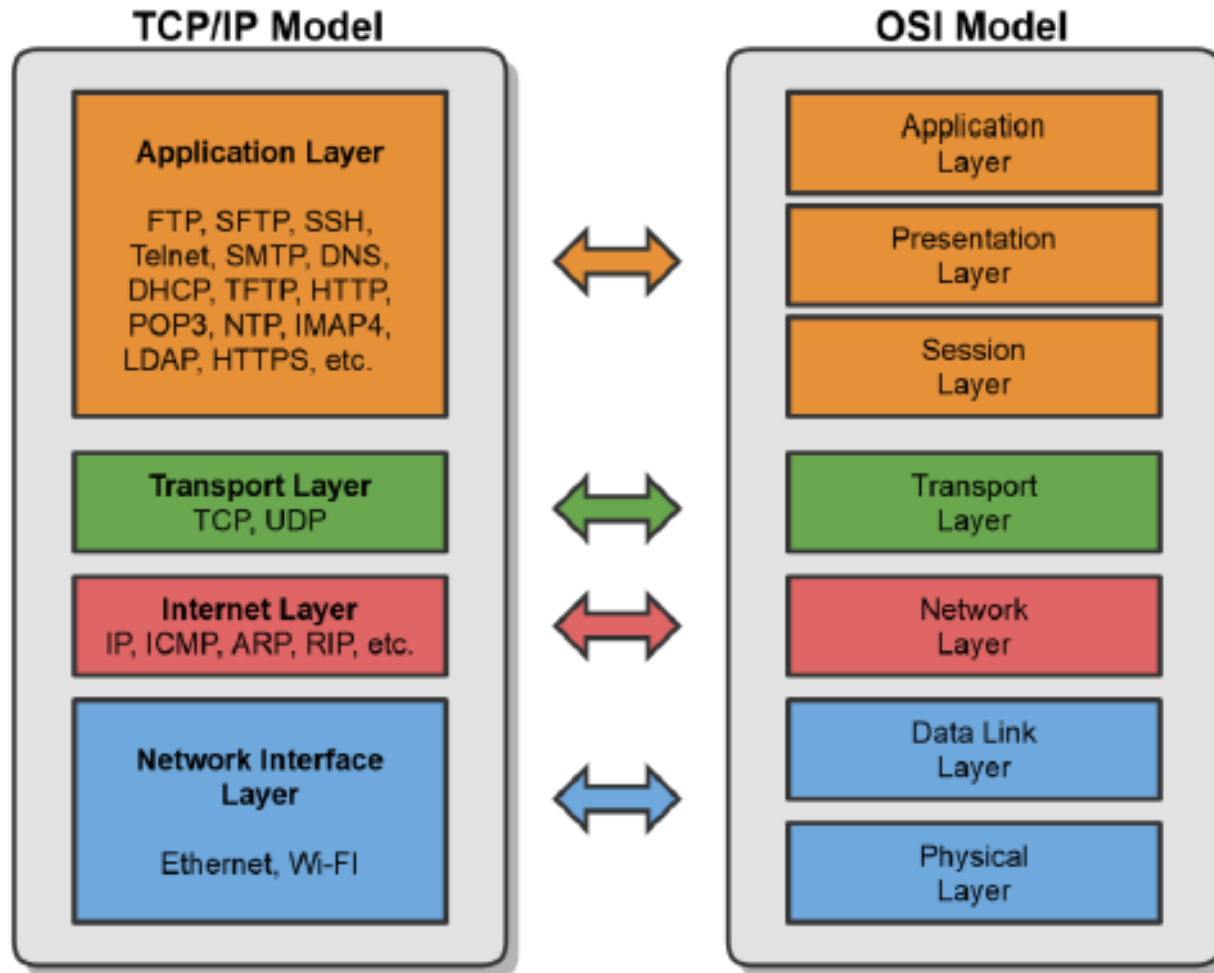
Layer	Protocols
Application	FTP, TFTP, DNS, HTTP(S), TLS/SSL, SSH, POP3, IMAP4, NTP, Telnet, SMTP, SNMP
Transport	TCP, UDP and Ports
Internet	IP Addressing (Routing), ICMP, ARP
Network Interface	Ethernet, Token Ring

These protocols work together to provide communication, management, diagnostics, and troubleshooting for a TCP/IP network.



Comparison of OSI Model and TCP/IP Model

Comparison of OSI Model and TCP/IP Model



Similarities:

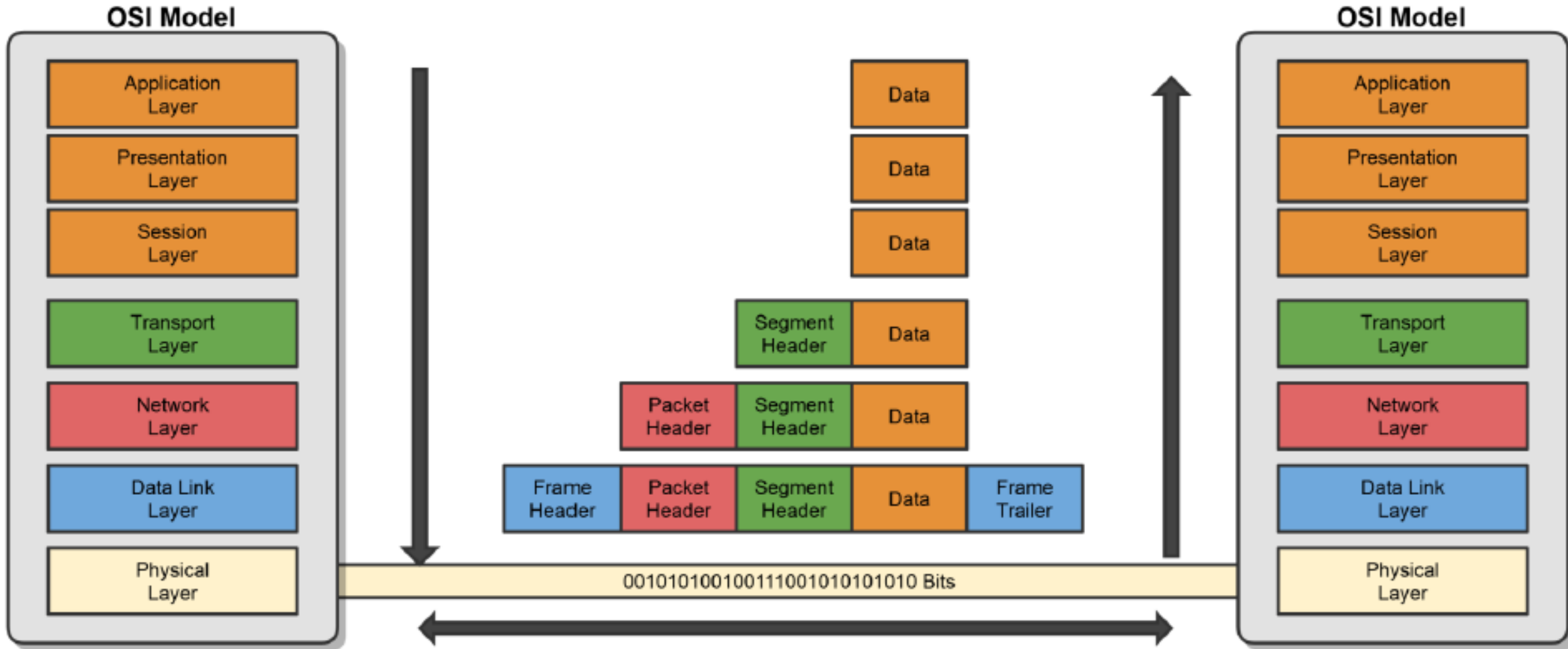
- Both have a layered structure.
- Both include an application layer.
- Data progresses through the layers in an end-to-end connected manner.

Comparison of OSI Model and TCP/IP Model

Differences:

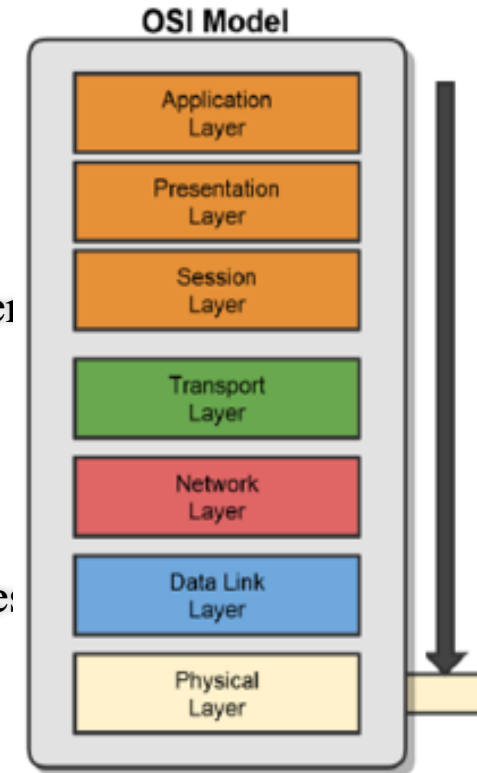
- TCP-IP is a protocol designed for applications.
- The Internet is built on the TCP-IP model.
- TCP-IP has fewer layers and appears simpler.
- In the OSI model, the roles of the layers are clearly defined, which makes it harder to place a new protocol within these layers when developed.
- TCP-IP is more flexible in this regard, allowing new protocols to be easily placed into the desired layer.

OSI Encapsulation & De-Encapsulation



Encapsulation Process (Sending Data):

1. **Application Layer:** Data originates from an application.
2. **Presentation Layer:** Data is translated, encrypted, or compressed as needed.
3. **Session Layer:** Data is marked for session management, ensuring correct communication.
4. **Transport Layer:** Data is segmented, and a transport header (segment header) is added. This header contains information such as port numbers and sequence numbers.
5. **Network Layer:** Segments are encapsulated into packets, and a network header (packet header) is added. This header contains logical addressing information like IP addresses.
6. **Data Link Layer:** Packets are framed, and a frame header and trailer are added. The header includes MAC addresses, and the trailer typically includes error-checking information.
7. **Physical Layer:** The frame is converted into bits and transmitted over the physical medium.



- Application Layer (Data):** The application generates data.
- Transport Layer (Segment):** Data is segmented and a transport header is added, creating a segment.
- Network Layer (Packet):** The segment is encapsulated with a network header, creating a packet.
- Data Link Layer (Frame):** The packet is encapsulated with a frame header and trailer, creating a frame.
- Physical Layer (Bits):** The frame is converted to bits for transmission over the physical medium.

Example of Encapsulation

1. Data (Application Layer):

```
"Hello, World!"
```

2. Segment (Transport Layer):

```
[Transport Header] "Hello, World!"
```

3. Packet (Network Layer):

```
[Network Header] [Transport Header] "Hello, World!"
```

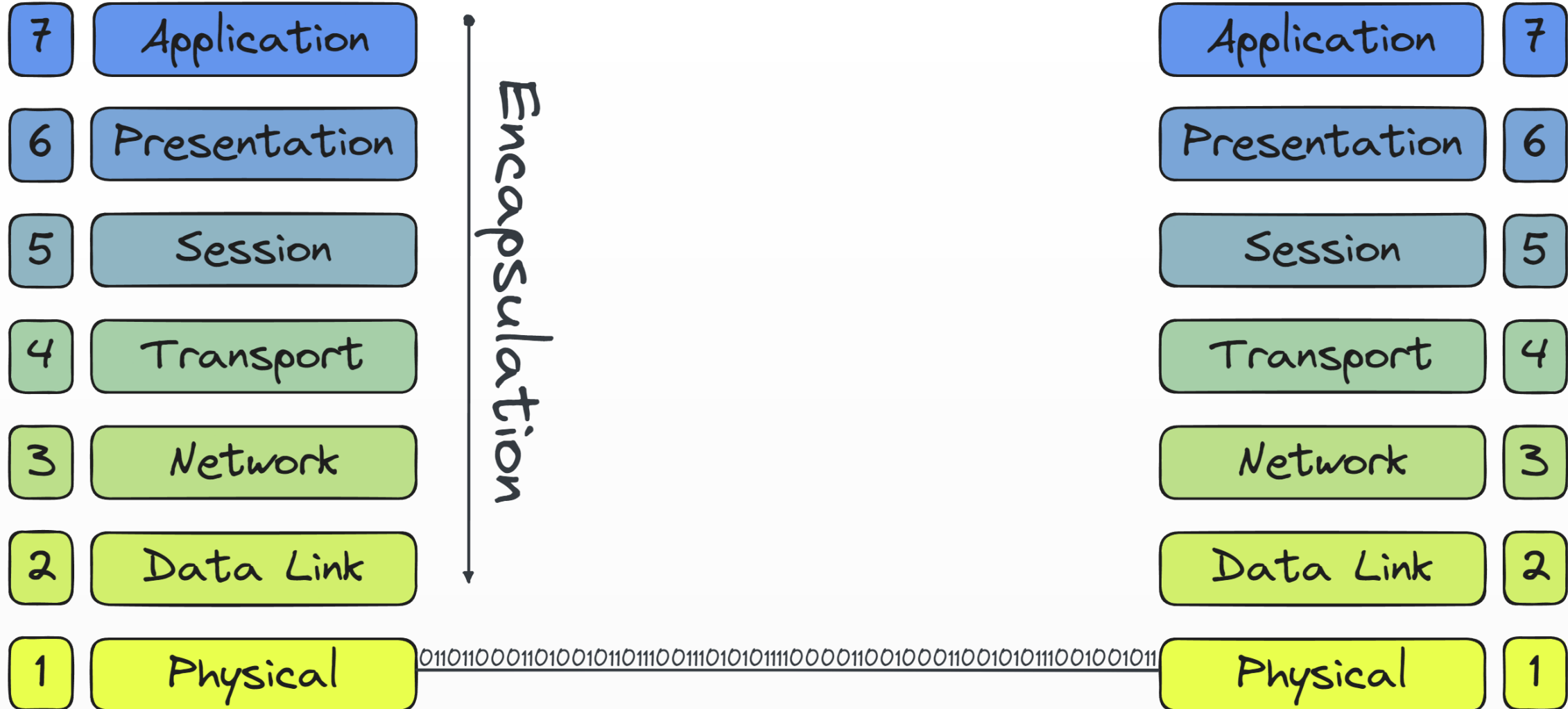
4. Frame (Data Link Layer):

```
[Frame Header] [Network Header] [Transport Header] "Hello, World!" [Frame Trailer]
```

5. Bits (Physical Layer):

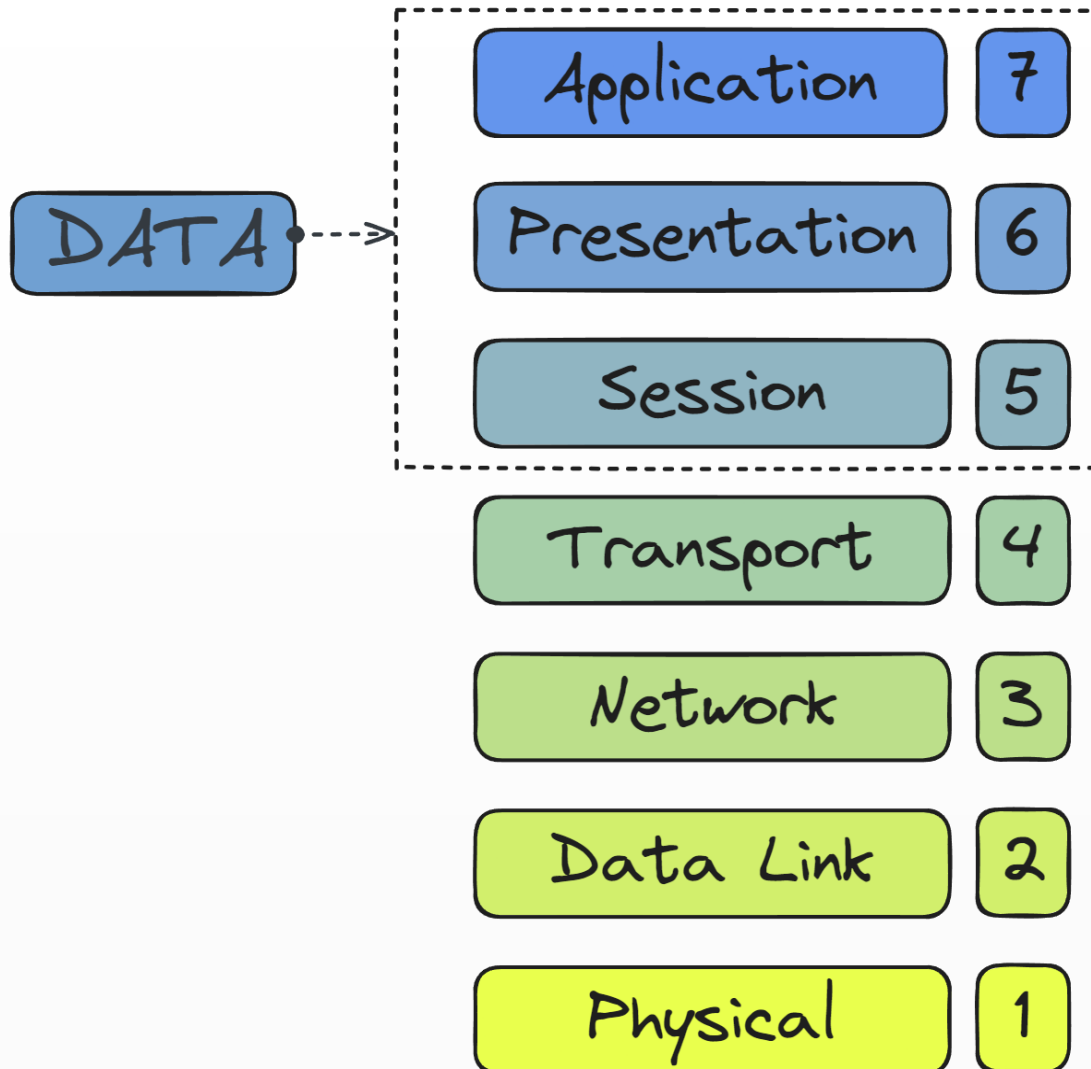
```
01010100 01101000 01100101 00100000 01000100 01100001 01110100 01100001...
```

Encapsulation



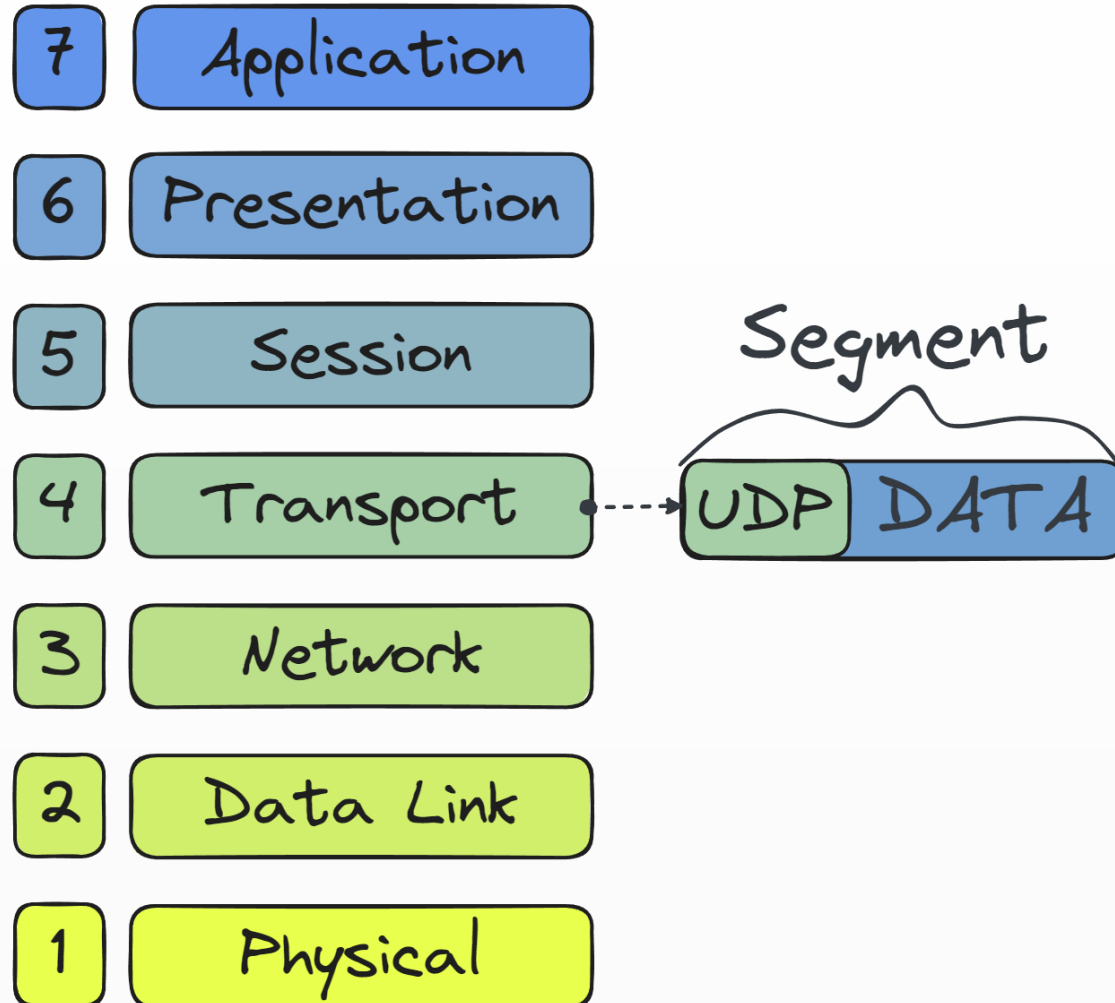
Encapsulation

OSI Model



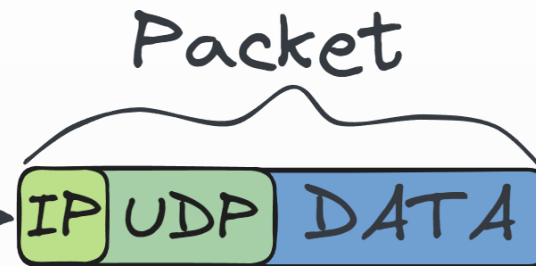
Encapsulation

OSI Model



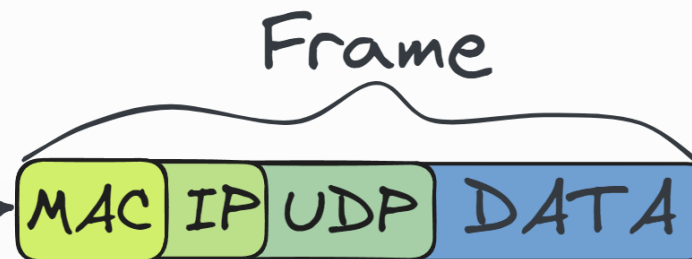
Encapsulation

OSI Model



Encapsulation

OSI Model



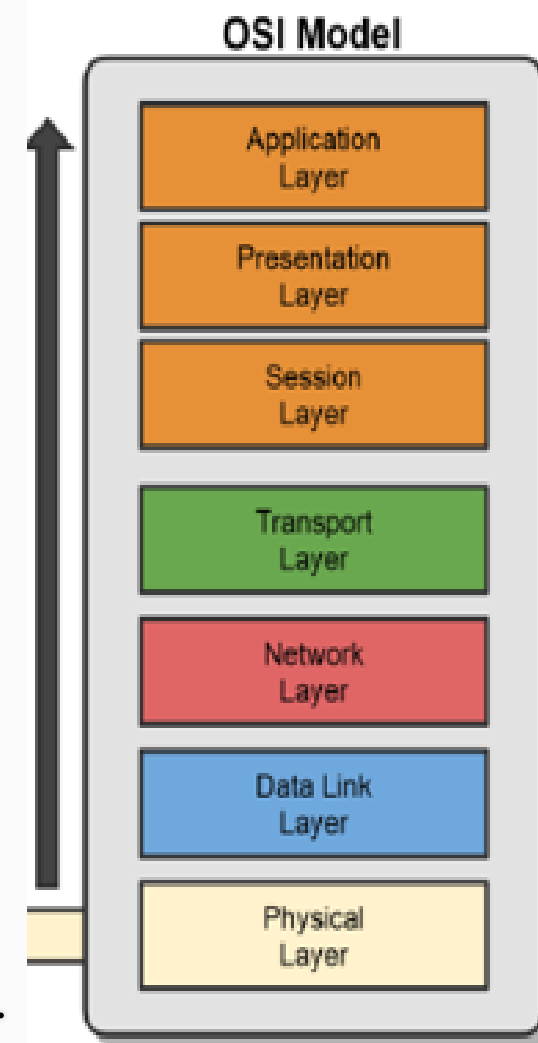
Encapsulation

OSI Model

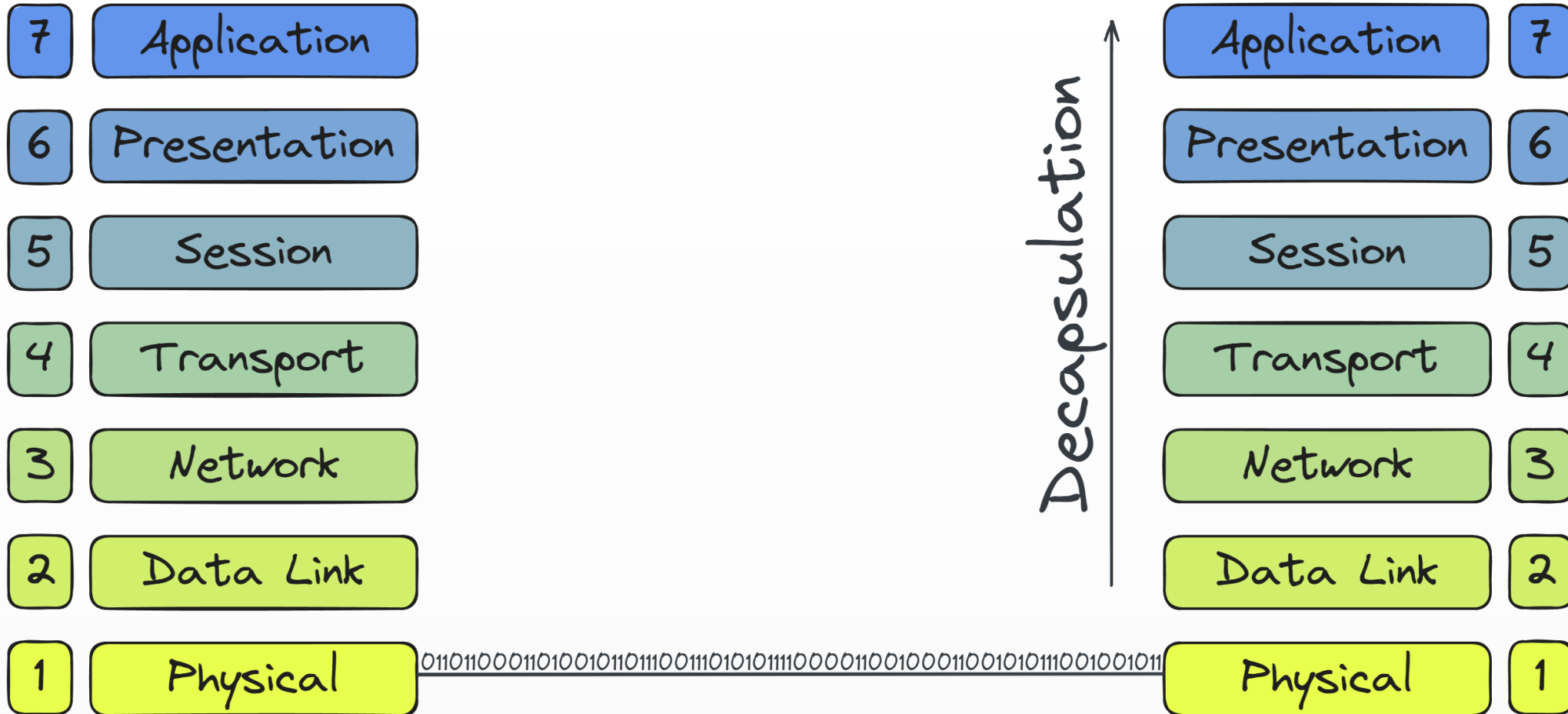


De-Encapsulation Process (Receiving Data):

1. **Physical Layer:** Bits are received and converted back into a frame.
2. **Data Link Layer:** The frame is checked for errors, and the frame header and trailer are removed, leaving the packet.
3. **Network Layer:** The packet header is removed, leaving the segment.
4. **Transport Layer:** The segment header is removed, reassembling the data.
5. **Session Layer:** Session information is used to re-establish the session.
6. **Presentation Layer:** Data is translated, decrypted, or decompressed as needed.
7. **Application Layer:** The data is presented to the application for use.

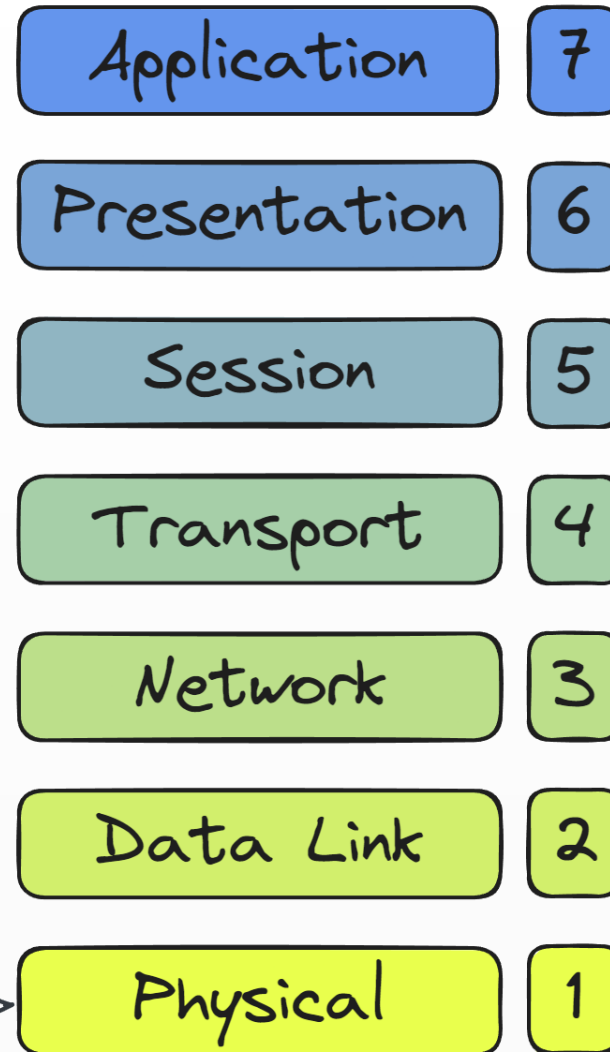


Encapsulation Decapsulation

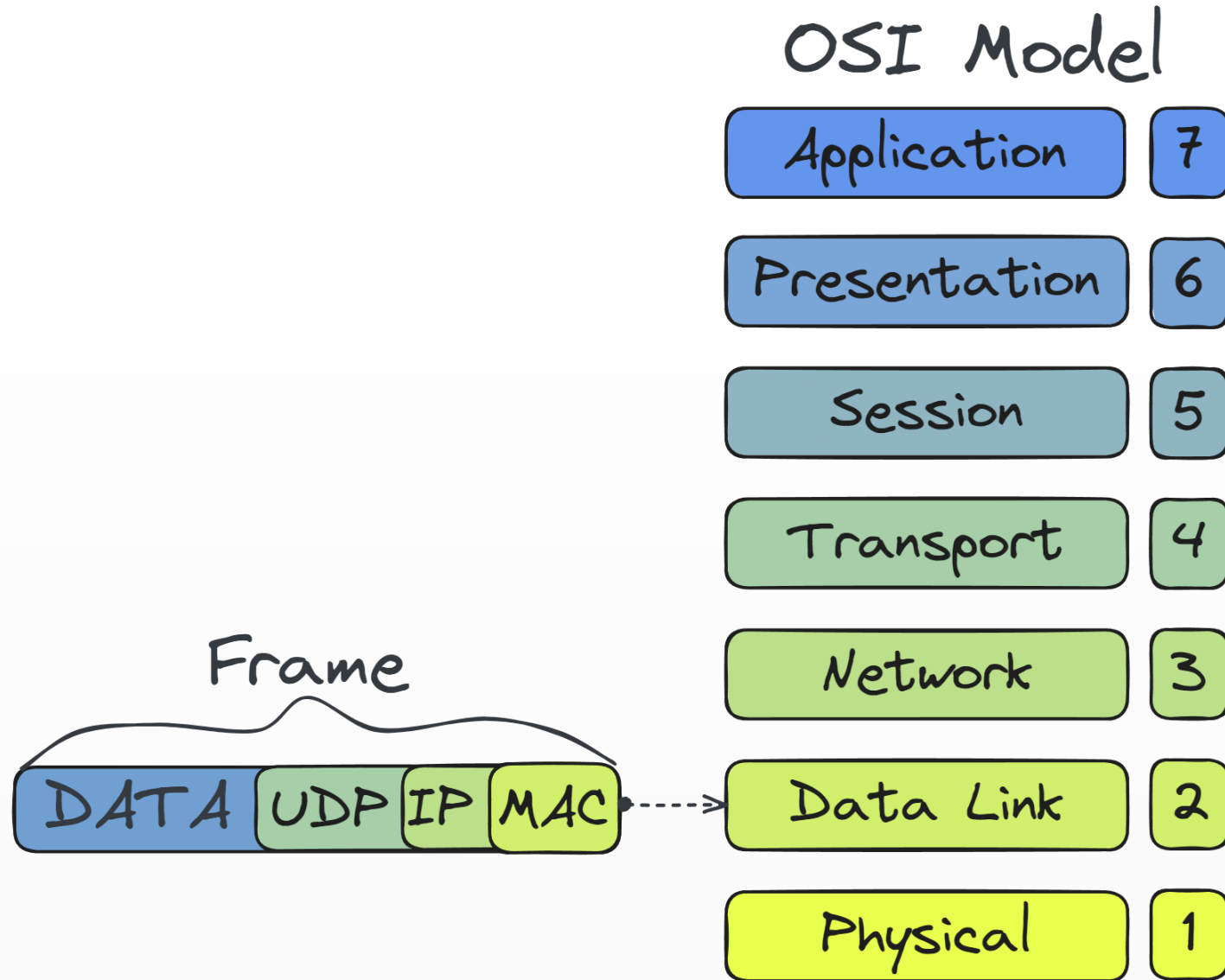


Decapsulation

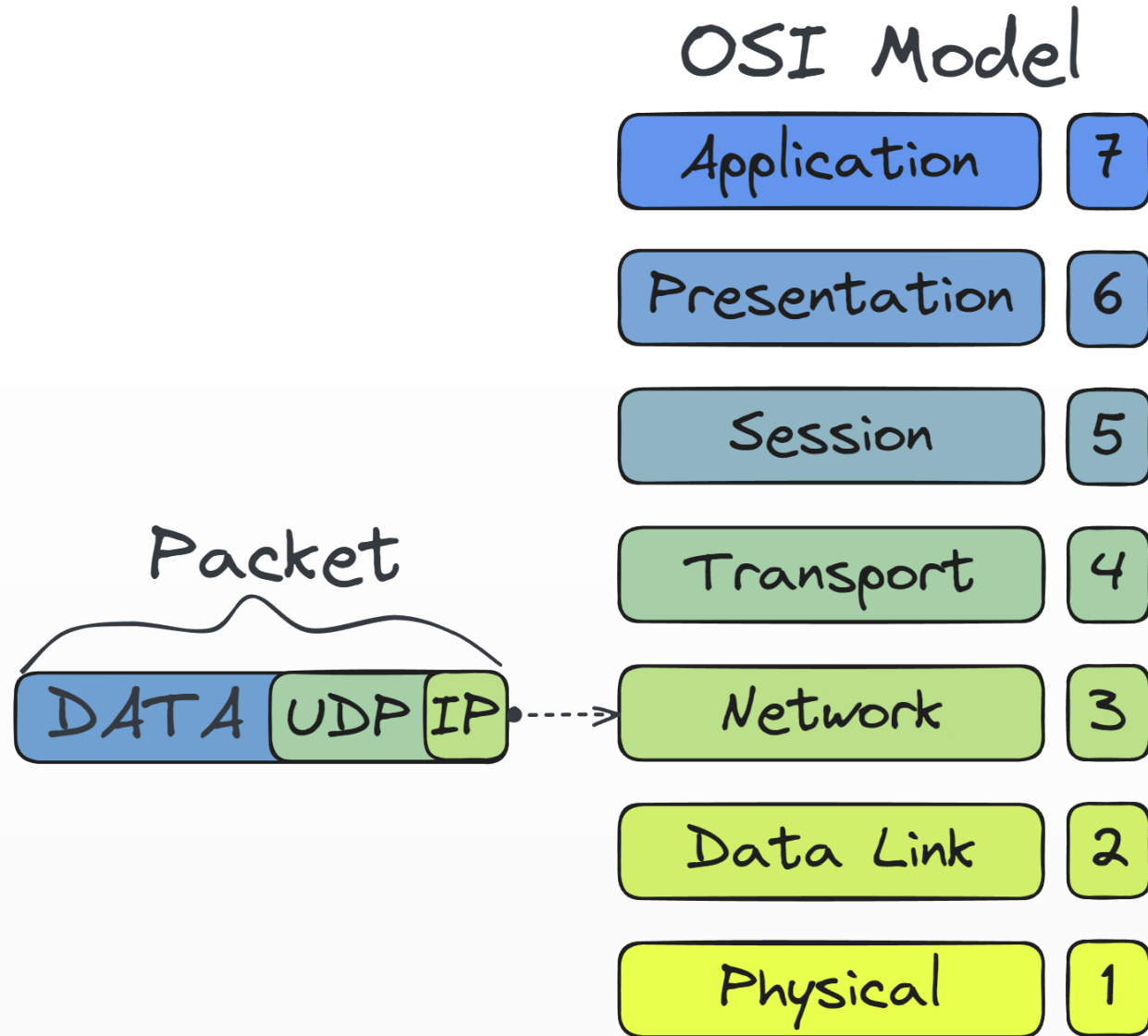
OSI Model



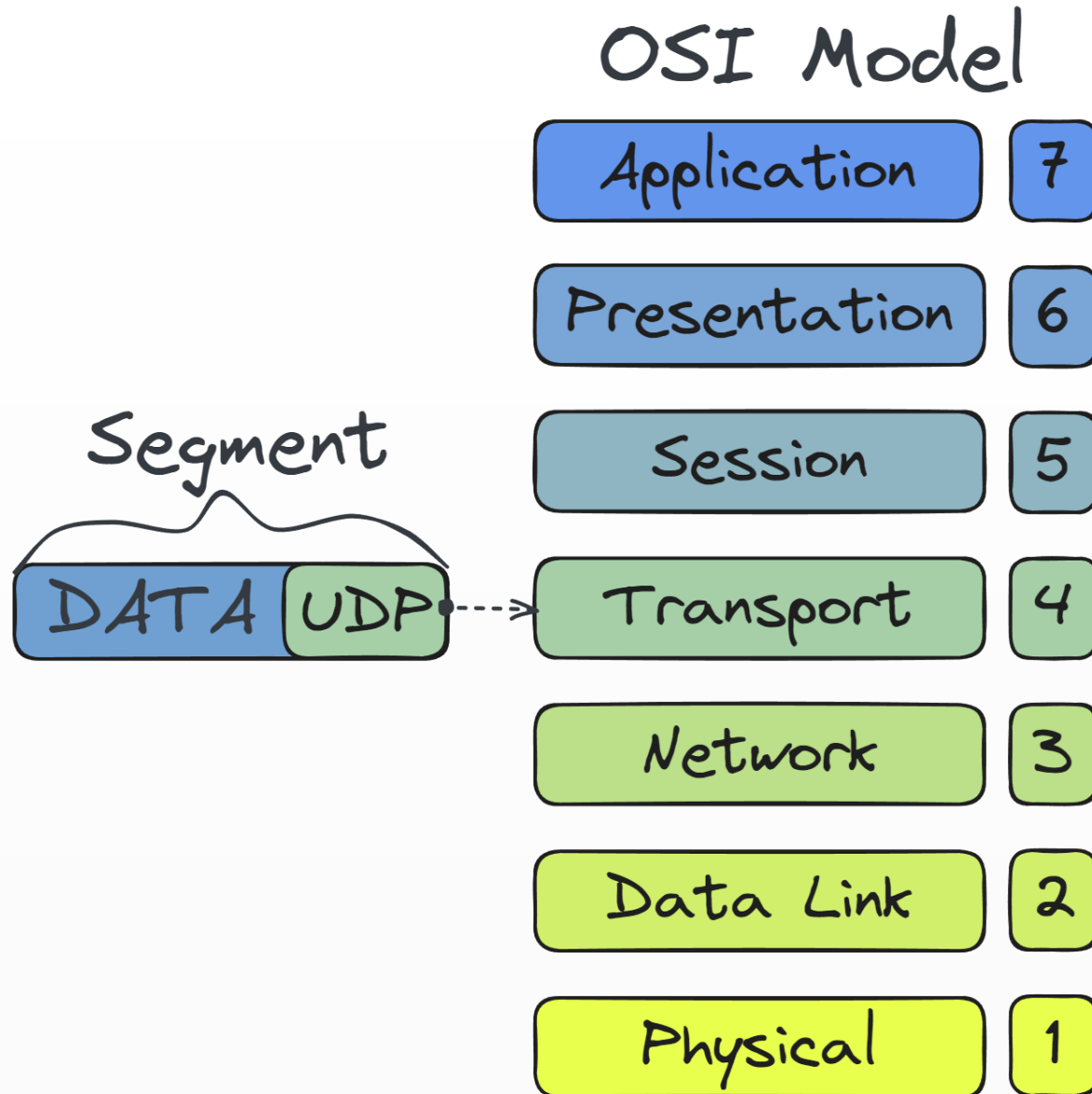
Decapsulation



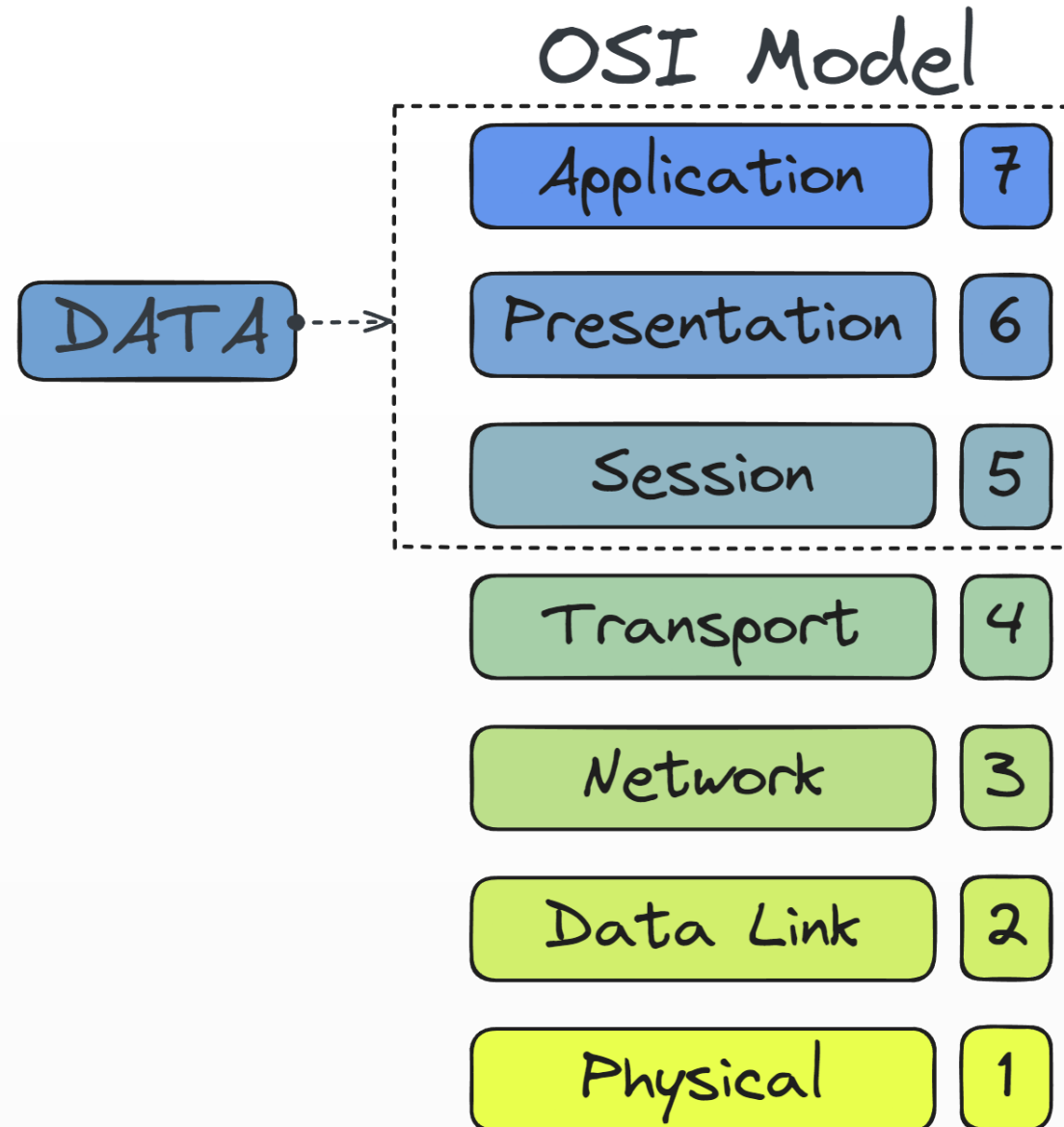
Decapsulation



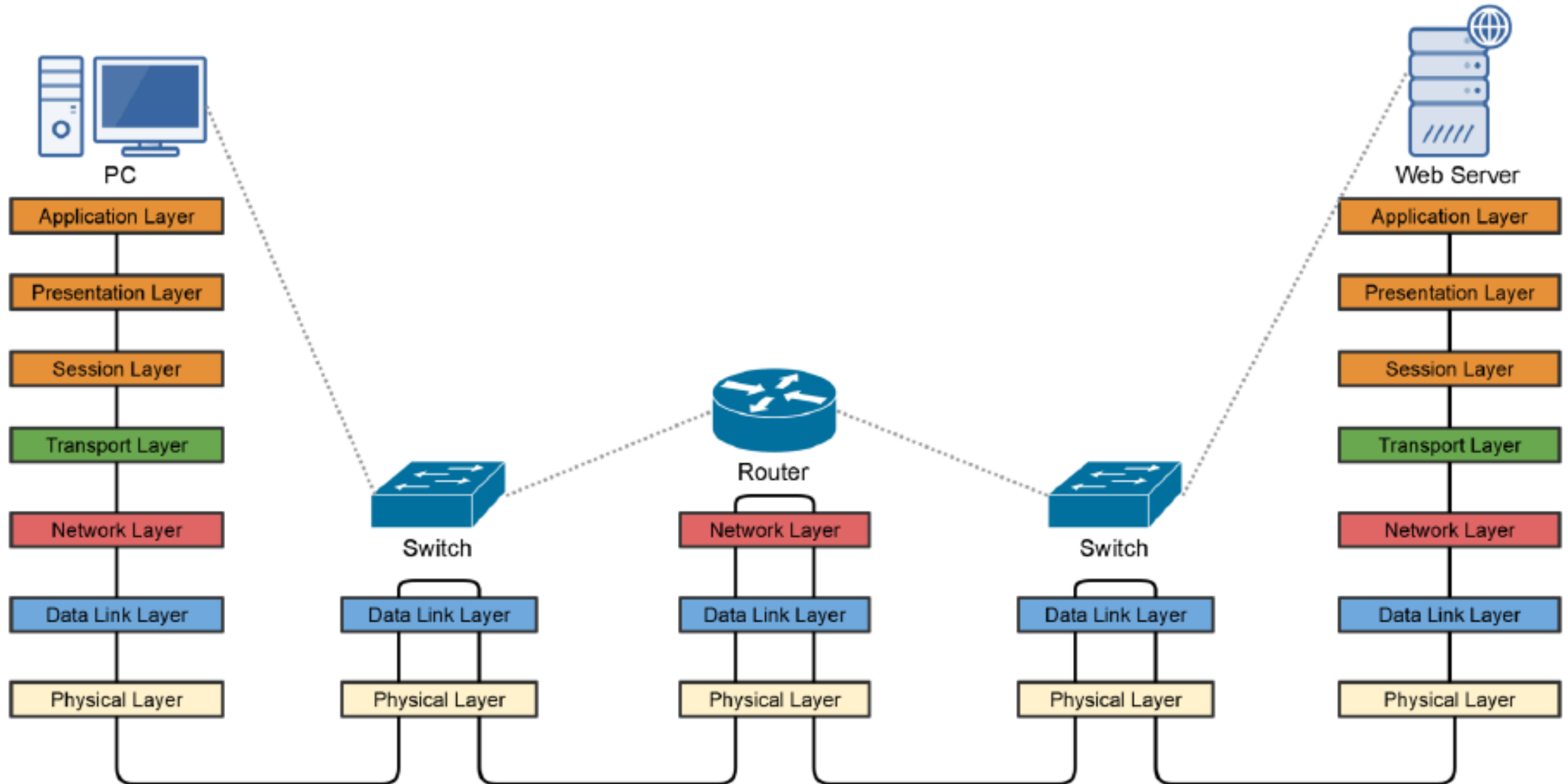
Decapsulation



Decapsulation



OSI Encapsulation & De-Encapsulation





Some Terms and Definitions

node

A connection point or end point for the transmission of the data

Hub, switch

These two nodes connect computers or other network devices. Hub broadcasts data to every computer, switch broadcasts to specific ones. Switch is smart.

NOS

Network Operating System

workstation

A special **computer** designed for technical or scientific applications. Intended primarily to be used by **one person at a time**, they are commonly connected to a **local area network**.

MAC address

Identification number of a device on the network

segment

Dividing network devices into groups

Packet-switching

Sending data as chunks, data broken into packets for a faster and secure communication.

server

A **server** is a **computer** or system that provides resources, data, services, or programs to other computers

client

Any device that makes request to servers

multimode fiber

fast but short distance fiber cable

proxy

Acts like gateway between user and Internet
Caching
Administrative control
Security

gateway

A node between 2 networks, makes them understand each other

Firewall

Network security device that controls incoming and outgoing network traffic

IP address

A unique number that defines a computer on the network

Router

A router connects 2 or more networks. It controls network traffic by forwarding data packets to the correct address.

Address Resolution Protocol (ARP)

- Resolves IP address to MAC Addresses
- Finds the hardware address of a host from a know IP address
 - And vice versa (RARP)

ARP Command: arp -a

```
Command Prompt
Microsoft Windows [Version 10.0.19042.985]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Alton>arp -a

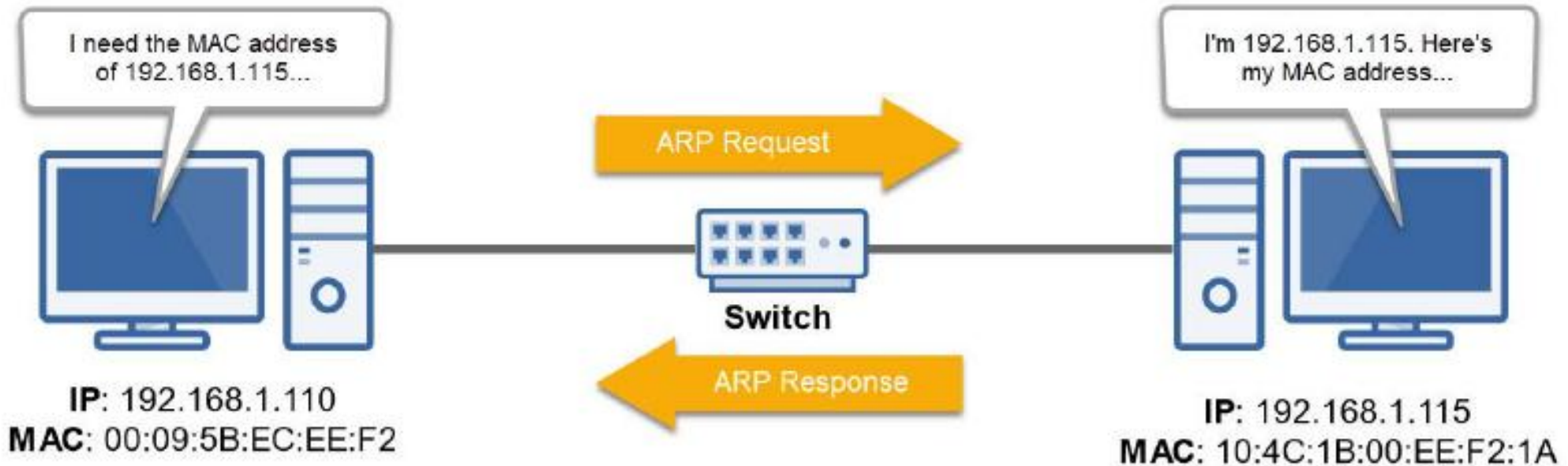
Interface: 192.168.0.132 --- 0xe
 Internet Address      Physical Address      Type
 192.168.0.1           2c-fd-a1-a2-74-c0     dynamic
 192.168.0.5           00-90-a9-db-c1-a3     dynamic
 192.168.0.10          00-11-32-e2-ce-58     dynamic
 192.168.0.15          00-11-32-d0-b6-9f     dynamic
 192.168.0.62          10-98-c3-dc-f4-4a     dynamic
 192.168.0.76          ac-ae-19-03-b3-e6     dynamic
 192.168.0.186         82-07-b3-9c-ef-ab     dynamic
 192.168.0.199         0c-47-c9-33-92-68     dynamic
```

```
root@kali: ~# arp -a
_gateway (10.0.2.1) at 52:54:00:12:35:00 [ether] on eth0
root@kali: ~#
```

```
alton — -bash — 68x7

Last login: Thu May 13 14:25:01 on console
[Altons-iMac:~ alton$ arp -a
? (10.0.2.2) at 52:54:0:12:35:2 on en0 ifscope [ethernet]
? (10.0.2.255) at ff:ff:ff:ff:ff:ff on en0 ifscope [ethernet]
? (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]
Altons-iMac:~ alton$
```

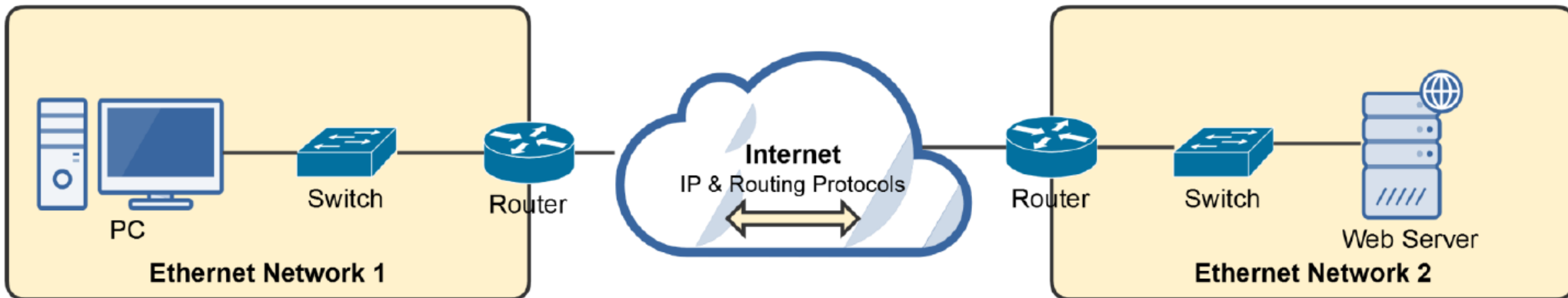
ARP Diagram



If a computer knows a device's IP address but not its MAC address, it'll send a **broadcast** message to all devices on the LAN asking which device is assigned that MAC address.

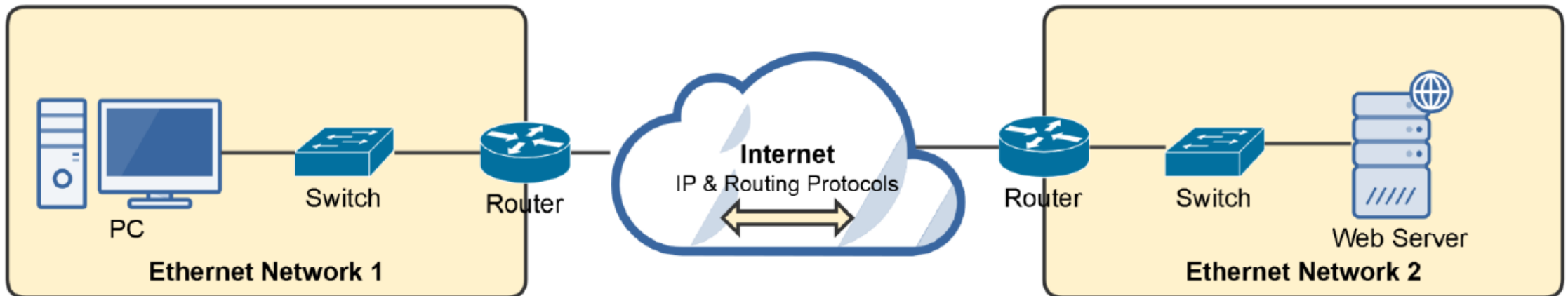
The Internet Protocol (IP)

- An OSI Layer 3 protocol that defines routing and logical addressing of packets that allow data to traverse WANs and the Internet.
- It specifies the formatting of packets and the logical addressing schema
 - **IP addresses:** IPv4 and IPv6
- Its job is to connect different OSI Layer 2 (switched) networks together.
- Provides end-to-end connectivity from one Layer 2 network to another via routers.



The Internet Protocol (IP)

- It's connectionless and, therefore, unreliable (similar to UDP).
 - No continued connection.
- Each packet sent is independent of each other packet.
 - TCP and other protocols provide a means to reassemble them properly.
 - Packets don't always follow the same path to their destination.
 - They're sent via the most efficient route.
- Doesn't provide any error recovery or sequencing functionality.
 - That's the job of other protocols.



Internet Control Message Protocol (ICMP)

- OSI Layer 3 Internet Protocol (IP) companion “error reporting” protocol within the TCP/IP suite of protocols.
- Just like IP, it’s connectionless.
- Used to generate error messages to the source IP address when network issues prevent the delivery of a packet.
- Typically used by routers to report packet delivery issues, and, most importantly, it can report errors but not correct them.
- Commonly used by IT administrators to troubleshoot network connections with command-line utilities, including ping, pathping, and traceroute.
- For IPv6, it is also used for:
 - Neighbor Solicitation and Advertisement Messages (Similar to ARP)
 - Router Solicitation and Advertisement Messages

(Some) ICMP Message Types

- **Echo Request, Echo Reply:** Tests destination accessibility and status. A host sends an *Echo Request* and listens for a corresponding *Echo Reply*. Commonly done using the **ping** command.
- **Destination Unreachable:** Sent by a router when it can't deliver an IP packet.
- **Source Quench:** Sent by a host or router if it's receiving too much data than it can handle. The message requests that the source reduces its rate of data transmission.
- **Redirect Message:** Sent by a router if it receives a packet that should have been sent to a different router. The message includes the IP address to which future packets should be sent and is used to optimize the routing.
- **Time Exceeded:** Sent by a router if a packet has reached the maximum limit of routers through which it can travel.
- **Router Advertisement, Router Solicitation (IPv6):** Allow hosts to discover the existence of routers. Routers periodically multicast their IP addresses via *Router Advertisement* messages. Hosts may also request a router IP address by broadcasting a *Router Solicitation* message, then wait for a router to reply with a *Router Advertisement*.

Understanding Protocols, Ports, and Sockets

Protocols

- Computers communicate with each other with network protocols.
- Protocols are rules governing how machines exchange data and enable effective communication.
- In an operating system (OS), a protocol runs as a process or service.

Ports

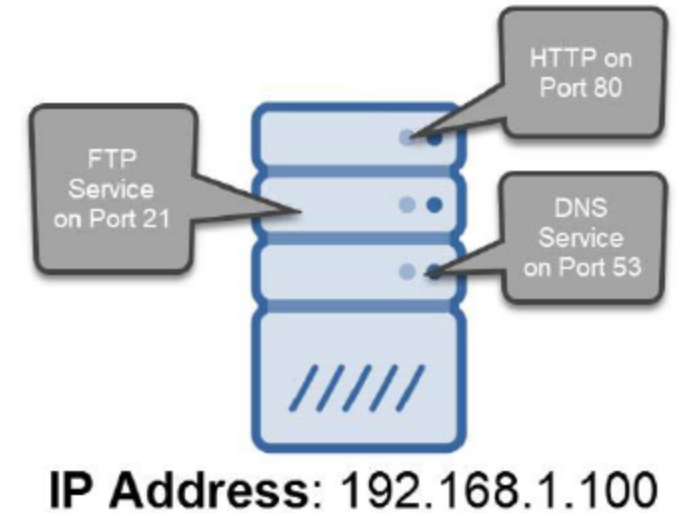
- Ports are logical constructs that bind a unique port number to a protocol process or service.

Sockets

- Sockets are a combination of an IP address and a port number, for example, 192.168.1.1:80.

Why We Need Ports and Sockets

- Computers require ports because of network application multitasking.
- Because a computer may have only one IP address, it needs ports to differentiate network protocols and services running on it.
- TCP/IP has 65,536 ports available



Port Type	Port Numbers	Description
Well Known Ports	0 – 1023	Assigned to well-known protocols.
Registered Ports	1024 – 49,151	Registered to specific protocols.
Dynamic Ports	49,152 – 65,535	Not registered and used for any purpose.

Protocols & Port Numbers

Service, Protocol, or Application	Port Number(s)	TCP or UDP
FTP (File Transfer Protocol)	20, 21	TCP
Secure FTP (SFTP)	22	TCP
SSH (Secure Shell Protocol)	22	TCP
Telnet	23	TCP
SMTP (Simple Mail Transfer Protocol)	25	TCP
DNS (Domain Name System)	53	UDP
DHCP (Dynamic Host Configuration Protocol)	67, 68	UDP
TFTP (Trivial File Transfer Protocol)	69	UDP
HTTP (Hypertext Transfer Protocol)	80	TCP
POP3 (Post Office Protocol version 3)	110	TCP

Protocols & Port Numbers

Service, Protocol, or Application	Port Number(s)	TCP or UDP
NTP (Network Time Protocol)	123	UDP
IMAP4 (Internet Message Access Protocol version 4)	143	TCP
SNMP (Simple Network Management Protocol)	161	UDP
LDAP (Lightweight Directory Access Protocol)	389	TCP
HTTPS (Hypertext Transfer Protocol Secure)	443	TCP
Server Message Block (SMB)	445	TCP
LDAPS (Lightweight Directory Access Protocol Secure)	636	TCP
RDP (Remote Desktop Protocol)	3389	TCP
ITU Telecommunication Standardization Sector A/V Recommendation (H.323)	1720	TCP
Session Initiation Protocol (SIP)	5060, 5061	TCP

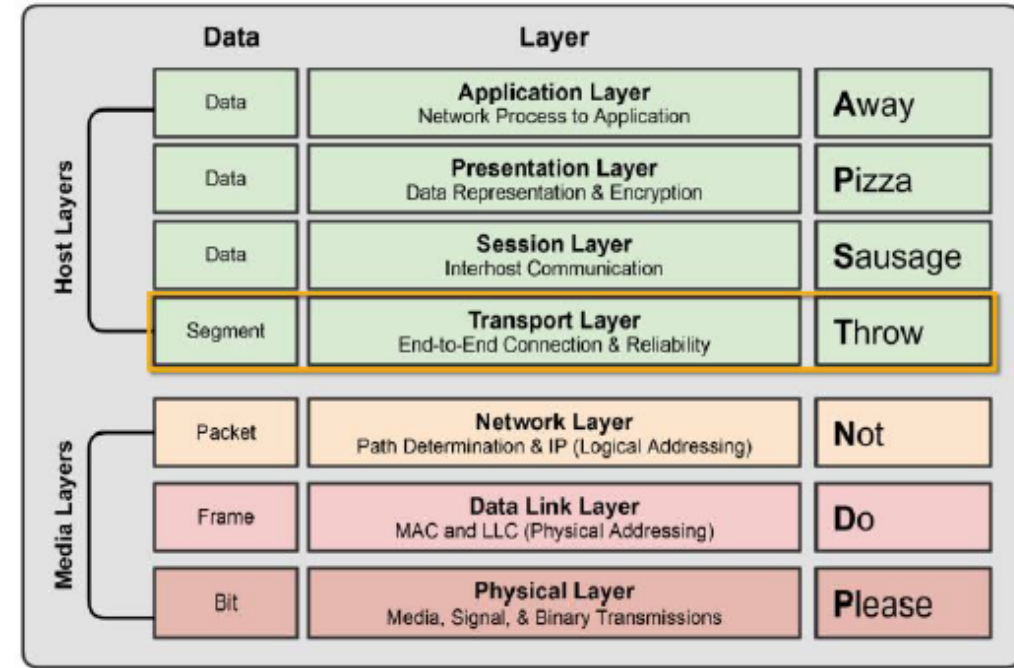
TCP vs. UDP

Transport Layer Protocols

- **TCP** (Transmission Control Protocol): Connection-Oriented
- **UDP** (User Datagram Protocol): Connectionless

TCP is the most widely used Transport Layer protocol because it is connection-oriented, which provides packet delivery reliability, i.e., guaranteed delivery.

UDP, being connectionless, is considered to be unreliable; however, it is more lightweight than TCP and often used for streaming or real-time data.

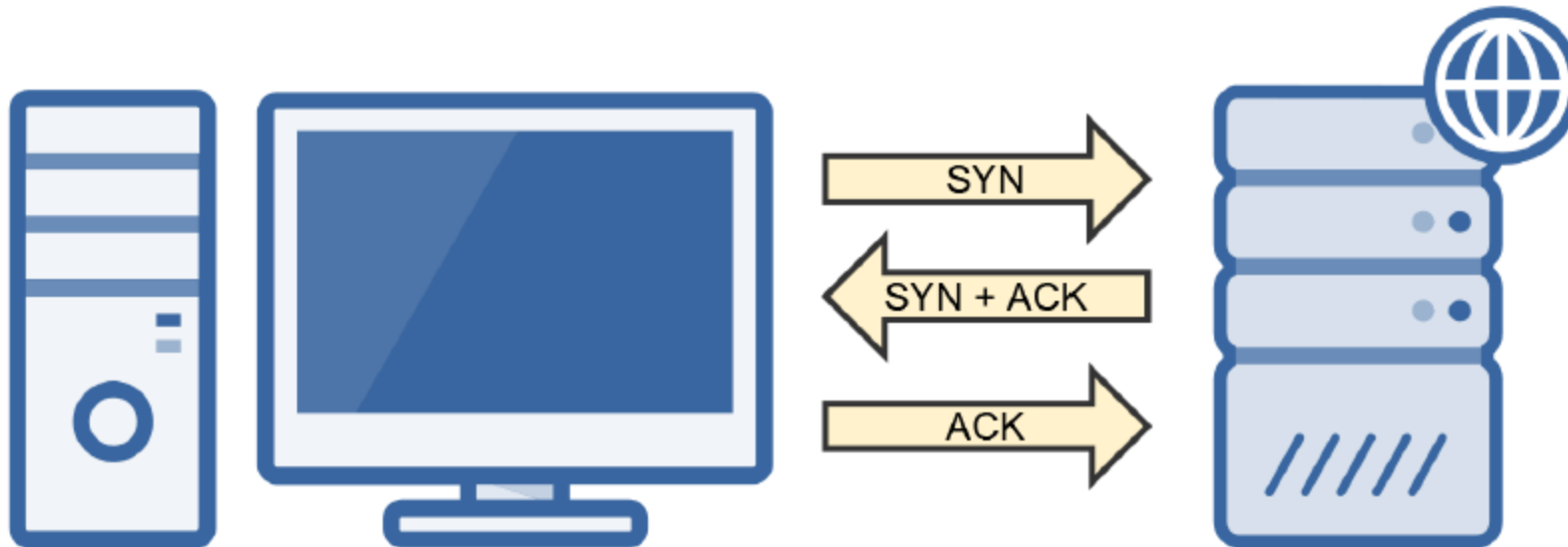


TCP Reliability

- TCP utilizes the following features to ensure reliable delivery of data.
 - **3-Way Handshake** creates a virtual connection between the source and destination before data is sent.
 - **Acknowledgment** is required before the next segment is sent.
 - **Checksum** that detects corrupted data.
 - **Sequence Numbers** that detect missing data and reassemble them in the correct order.
 - **Retransmission** that will retransmit lost or corrupt data.
- **Note:** TCP header is 20 bytes in size, whereas the UDP header is only 8 bytes.

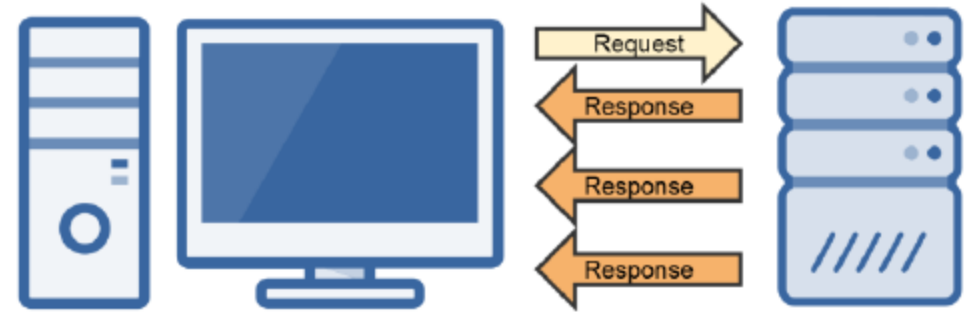
TCP Three-Way Handshake

- A connection must be established before data is transmitted, called the three-way handshake.
 - $\text{SYN} \rightarrow \text{SYN} / \text{ACK} \rightarrow \text{ACK}$
- Creates a Virtual Connection Between 2 Devices



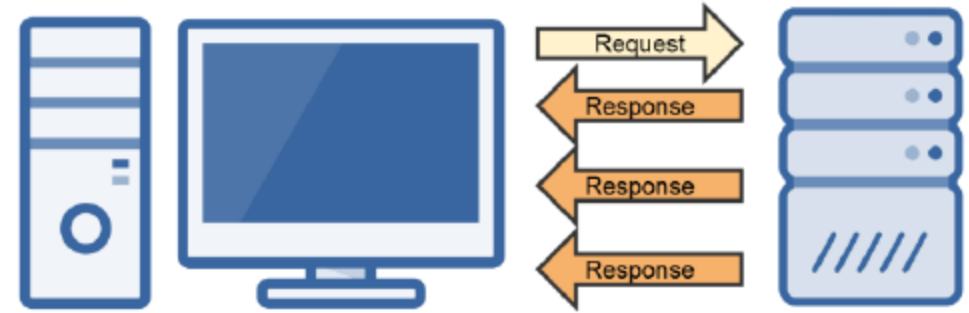
“Best Effort” UDP

- A scaled-down, economic version of TCP
 - Connectionless & Unreliable
 - No Data Retransmissions
 - “Best Effort”
- Faster than TCP
 - Smaller Header & Connectionless
- Primarily used for protocols that favor:
 - Low-Latency, i.e., Faster Speeds
 - Can Tolerate Data Loss



“Best Effort” UDP

- Example UDP Use-Cases
 - VoIP Phone Calls
 - Live Video Streams
 - Live Audio Streams
 - Online Gaming
 - Certain Network Management Protocols
 - DNS
 - DHCP
 - NTP



Application Layer Management Protocols

- Domain Name System (DNS)
- Dynamic Host Configuration Protocol (DHCP)
- Network Time Protocol (NTP)
- Simple Network Management Protocol (SNMP)
- Lightweight Directory Access Protocol (LDAP)
- LDAP Secure (LDAPS)
- Server Message Block (SMB)

Domain Name System (DNS)

Port: 53 Transport Layer Protocol: UDP

- Protocol that is used to resolve a domain name to its corresponding IP address
 - InstructorAlton.com → 162.0.232.236
- Uses TCP port 53 by default
- We'll be discussing DNS in detail in the **DNS Network Services** section of this course:
 - DNS Hierarchy
 - DNS Record Types
 - Name Resolution

Dynamic Host Configuration Protocol (DHCP)

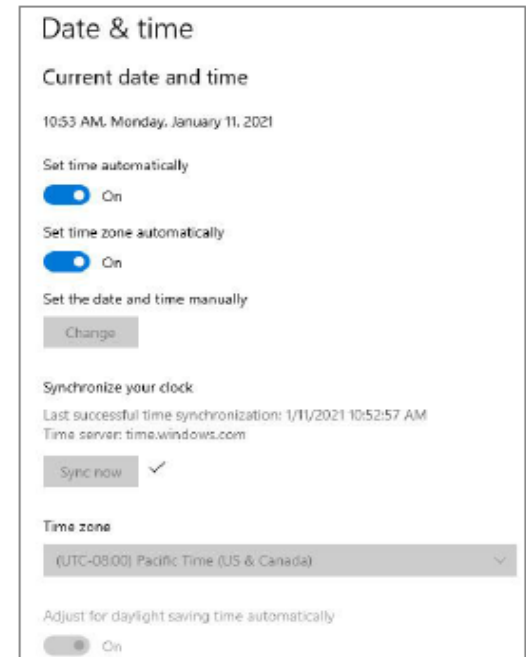
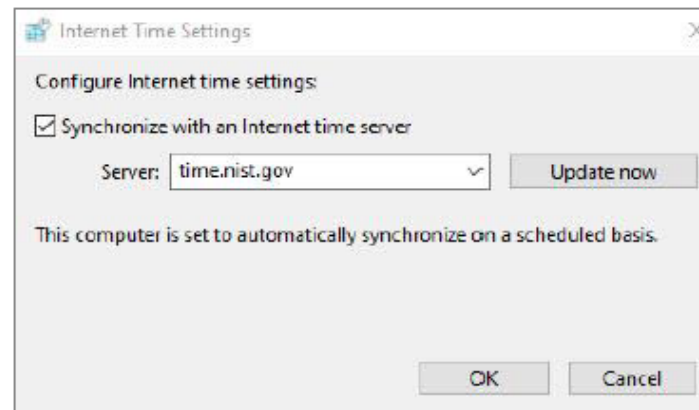
Ports: 67, 68 Transport Layer Protocol: UDP

- Protocol that automatically assigns IP address configurations to devices on a network:
 - IP Address
 - Subnet Mask
 - Default Gateway
 - DNS Server
- We'll be discussing how DHCP works in detail in the **Assigning IP Addresses** section of this course
- Uses two UDP ports 67 and 68 by default

Network Time Protocol (NTP)

Port: 123 Transport Layer Protocol: UDP

- Protocol that automatically synchronizes a system's time with a network time server.
 - Important for time-dependent network applications and protocols.
 - If a system is configured with the incorrect time, it may not be able to access network services.
 - Authentication will often fail if time isn't properly synchronized between devices.
- Uses TCP port 123 by default.



Simple Network Management Protocol (SNMP)

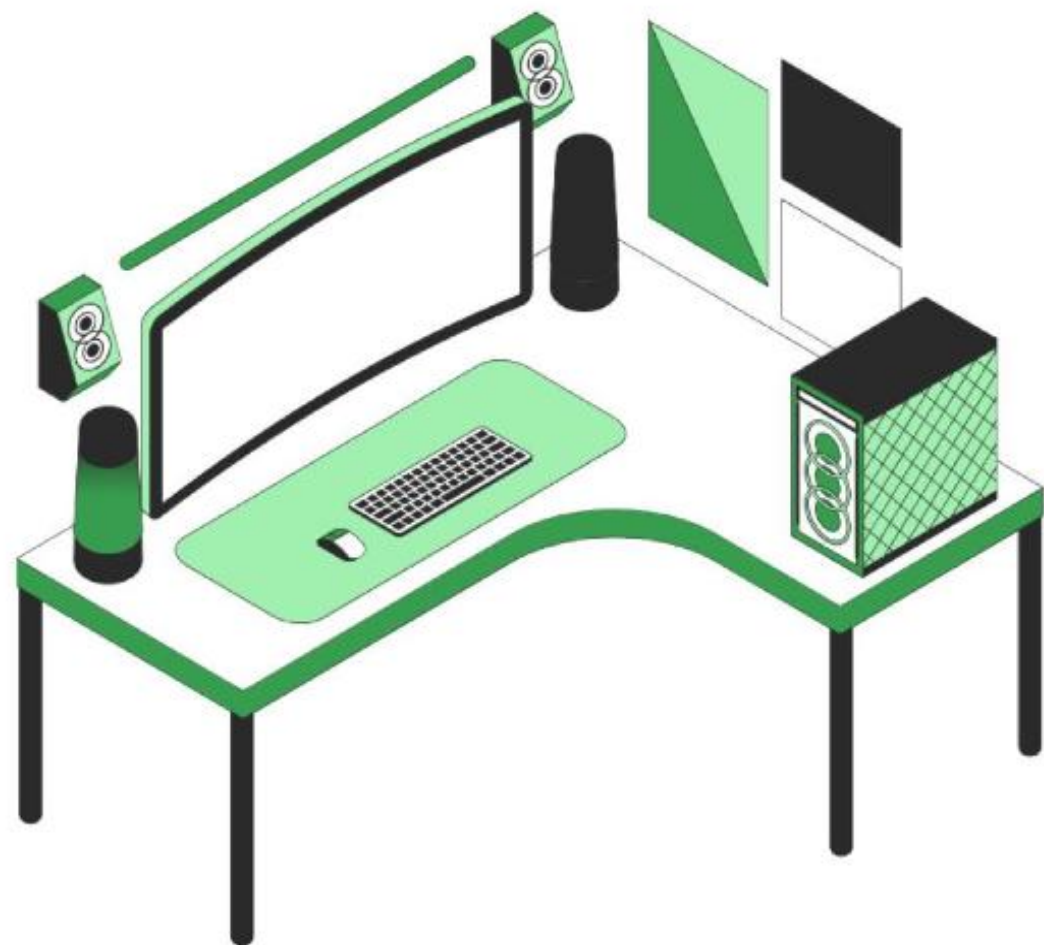
Port: 161 Transport Layer Protocol: TCP

- Protocol used to monitor and manage network devices
- Allows admins to monitor and manage network devices and traffic.
- Allows network devices to communicate information about their state:
 - Memory
 - CPU
 - Bandwidth
- Uses TCP port 161 by default

Server Message Block (SMB)

Port: 445 Transport Layer Protocol: TCP

- Network and file sharing protocol commonly used in Microsoft environments
- Allows systems to share their files and printers with other systems
- Uses TCP port 445 by default



Do you have any questions?

Send it to us! We hope you learned
something new.